30. Abstract Construction of Ideal Theory in Algebraic Number and Function Fields

Math. Ann. 96 (1927), pp. 26–61.

In what follows an abstract characterization is given of all those rings whose ideal theory agrees with the ideal theory of all integers of an algebraic number field: that is, whose ideals can be represented uniquely as products of powers of prime ideals. Among the best-known examples of such rings are also the ring of all integers of an algebraic function field in one indeterminate, and, more generally, in several indeterminates as soon as, following Kronecker, one restricts to ideals of highest dimension, i.e. passes to a certain quotient ring, the functional domain.

For the underlying commutative ring, the axioms completely equivalent to this ideal theory are the following:[1]

I. Divisor-chain condition. Every chain of ideals in which each ideal is a proper divisor of the preceding one terminates after finitely many steps; in other words, in every divisor chain of ideals there is an index from which on all ideals are equal.

II. Multiple-chain condition modulo every nonzero ideal. Every chain of ideals — all of which are divisors of a fixed ideal different from the zero ideal — in which each ideal is a proper multiple of the preceding one terminates after finitely many steps.

III. Existence of the identity element for multiplication.

IV. Ring without zero divisors.

V. Integral closedness in the quotient field. Every element of the quotient field that is integral with respect to the ring belongs to the ring.

From these axioms I develop, step by step, an increasingly restricted ideal theory up to the desired one; conversely, I show that all these axioms are in fact satisfied there.

From the divisor-chain condition I it follows, as I showed in *Ideal Theory* §4, that an ideal is representable as the least common multiple of finitely many primary ideals belonging to distinct prime ideals. I recall this proof briefly here (§6), both because it can be simplified by using the concept of the ideal quotient, and because I want to correct an oversight: the original proof uses, at one point, the not-assumed existence of the identity element of multiplication.[2]

The assumption of the multiple-chain condition has the effect (§7) that, in the residue class ring modulo every ideal different from the zero ideal, a prime ideal cannot have a proper divisor, except for the unit ideal containing all elements. Consequently the primary components of these ideals are determined uniquely, except for those belonging to the unit ideal. In a somewhat less sharp form this result is already found in Masazo Sono;[3] his assumption of the existence of a composition series in the residue class ring is equivalent to the "double chain condition" in this ring, as I show briefly at the end (§10). By double chain condition I mean the validity of both the divisor-chain and multiple-chain conditions without restriction to ideals different from the zero ideal.

If axiom III — existence of the identity element — is also satisfied, the unit ideal does not occur as an associated prime ideal. The primary components are therefore uniquely determined; they are pairwise coprime, and their least common multiple equals their product. Axioms I, II, III hold for all finite orders of an algebraic number field; already Dedekind (Dirichlet–Dedekind, *Zahlentheorie*, 3rd ed., 11th supplement,

[1] For the definition of the basic concepts of ideal theory, compare for instance E. Noether, *Idealtheorie in Ringbereichen*, Math. Ann. 83 (1921), pp. 24–66 (cited below as *Ideal Theory*). Incidentally, the most essential concepts are also formulated briefly in the present paper, especially in §§1, 4, and 5. For completeness, some material is included that is not needed for the immediate purposes of this paper.

[2] P. Urysohn called my attention to this oversight; my own alteration of the proof was somewhat more cumbersome than the one given here, which is due to E. Artin. He had already filled the gap for himself earlier, and, independently of me, had also observed the simplification with the ideal quotient. — I take a further simplification, which avoids the introduction of the "reduced representation," from a related question in W. Krull, *Algebraische Theorie der Ringe III*, Math. Ann. 92 (1924), pp. 181–213, Theorem I.

[3] Masazo Sono, *On the Reduction of Ideals*, Memoirs of the College of Science, Kyoto University, Series A, 7 (1924), pp. 191–204.

§172), without giving the proof in full detail, stated here the unique representation of an ideal as a product of pairwise coprime one-kind ideals (primary components).

From assumption IV — absence of zero divisors — it follows that the different powers of an ideal distinct from both the zero ideal and the unit ideal are all distinct, and that the quotient field exists. If finally axiom V of integral closedness in the quotient field is also satisfied, the primary ideals — uniquely determined because of IV — become powers of prime ideals, and the usual factorization theorem is obtained (§8). This latter proof rests on a consequence of integral closedness already drawn by Dedekind (3rd ed., §172) in the case of number fields: a genuinely fractional element can be represented as a quotient of integers in such a way that even the square of the numerator is not divisible by the denominator. In later presentations this consequence is replaced — likewise as a consequence of integral closedness — by the much less transparent generalized Gaussian theorem or by corresponding module theorems that say somewhat more; all of these are theorems whose application presupposes basis elements, whereas here one works directly with the ideals themselves.

In the converse direction (§9), the axioms distinct from axiom V follow almost directly; V follows from the proof that a genuinely fractional element can always be represented so that no power of the numerator is divisible by the denominator, which is therefore a strengthening of the consequence originally derived from V. This proof succeeds only after one draws from the ideal representation the usual conclusions: representation by principal ideals modulo a fixed ideal, and the theory of fractional ideals, i.e. of ideal quotients in the field. The fact that integral closedness follows from the representation by powers of prime ideals was also already known to Dedekind, as is shown by a remark in the 3rd edition, §172, p. 522, bottom.

Axioms I through V imply that the four decompositions which are generally distinct — into least pairwise coprime ideals, into mutually prime ideals, into largest primary components, and into irreducible ideals — coincide; conversely, from this coincidence and axioms I–IV, integral closedness follows. For rings with zero divisors, fulfillment of the remaining axioms — with V defined as integral closedness in the quotient ring — does not necessarily imply that the decompositions coincide, as the example of the residue class ring modulo certain ideals of a finite order shows (§9).

In §1 I sketch the theory of quantities integral with respect to a ring, up to Dedekind's consequence from integral closedness. In §2 I show how the chain conditions are transferred from the ring to modules consisting of linear forms, with this ring as coefficient and multiplier domain.[4] From this I draw (§3) the conclusion that, on passing to the integral elements of an algebraic extension field of the first kind, axioms I to IV remain valid for every order, while for the order consisting of all integral elements, V is also valid. This fact — of course familiar for the algebraic number field — permits the subordination of the function fields mentioned at the beginning; in particular, by the simple proof that the functional domain derived from integral polynomials in several indeterminates satisfies the axioms, Kronecker's ideal theory is fully subsumed.[5]

The ideal theory subsequently developed does not presuppose these §§2 and 3, which serve only for classification. In §4 and §5 the foundations independent of axioms I through V are given; this brings out more sharply than in the original foundation which parts of the theory are independent of finiteness assumptions.

§1. Theory of Integral Quantities

Let there be given a commutative ring[6] T without zero divisors, with identity element e for multiplication (axioms III and IV); in T a fixed subring R containing the identity element is distinguished. The concepts module, order, integral are throughout to refer to this fixed ring R, which later will be omitted from the

[4]This form of the module theorems is due to Prüfer (Neue Begründung der algebraischen Zahlentheorie, §2, Math. Ann. 94 (1924), pp. 198–243) and is more transparent than the customary transfer of the basis theorems.

[5]The difference appears only in the discriminant theorems, because the residue class ring of the functional domain modulo a prime number becomes an imperfect field, and hence extensions of the second kind can occur.

[6]As is known, a ring is defined once an equality relation is given in a set, and in addition two operations, addition and multiplication, are given satisfying the usual laws: addition is associative, commutative, and uniquely reversible; multiplication is associative — commutative when the ring is commutative — and distributive with respect to addition. If the underlying definition of equality is not set-theoretic identity, then, so that the same definition of equality is retained in every subsystem, such a subsystem (subring, module, ideal) must contain with any element all elements equal to it. Likewise, in every extension it must be assumed that the new definition of equality includes the old one. All concepts such as "finitely many elements," "unique assignment," etc. are meant in the sense of the equality definition; for example, "there is one and only one element" means that there is only one element up to equality. Incidentally, one can always pass from equality to set-theoretic identity by collecting equal elements into a class and introducing these classes as the new elements. The remarks given here on the definition of equality are due to R. Hölzer.

notation for brevity.[7] Elements of R will be denoted by Latin letters, elements of T in general by Greek letters.

1. A system M of elements of T is called, as usual, an R-module — briefly, a module — if, together with two elements α and β, it also contains their difference, and with α also $r\alpha$, where r is any element of R. M is said to be divisible by N,

$$M \equiv 0 \pmod{N},$$

and M is called a multiple, N a divisor, if every element of M is also contained in N. R-modules all of whose elements belong to R are called ideals in R.

Clearly the intersection — the least common multiple $[\ldots, M_i, \ldots]$ — of any number of modules from T is again a module; likewise T itself is a module. Thus the module derived from any system Σ of elements of T exists uniquely as the intersection of all modules containing Σ. The sum — the greatest common divisor $(\ldots, M_i, \ldots)$ — of any system of modules is the module derived from their union. The product MN of two modules is defined as the module derived from the elements $\mu\nu$, where μ runs through all elements of M and ν through all elements of N. The product therefore satisfies the associative and commutative laws, since these hold for the ring elements. The distributive law in T further gives, for modules, the distributive law

$$(\ldots, M_i, \ldots)N = (\ldots, M_iN, \ldots),$$

since by exactly this law in T both sides are the module derived from all elements $\mu_i\nu$.

Since R itself is a module, the condition that R is the multiplier domain can be expressed by $RM \equiv 0 \pmod{M}$. Since by hypothesis R contains the identity element, one also has $M \equiv 0 \pmod{RM}$, and hence $M = RM$.

A module M is called finite if there is at least one system Σ consisting of finitely many elements from which M is derived; the elements $\mu_1, \ldots, \mu_n$ of Σ are called a module basis of

$$M = (\mu_1, \ldots, \mu_n).$$

The sum and product of two — and hence of finitely many — finite modules are again finite, since they are derived from the union, respectively from the products, of the basis elements. The latter assertion follows from the representation $r_1\mu_1 + \cdots + r_n\mu_n$ of all elements μ of M by the basis and from the commutative law of multiplication in T.

2. An extension ring of R lying in T — hence one that contains all elements of R — is called an R-order, briefly an order. The intersection of arbitrarily many orders in T is again an order, and likewise T is an order; hence the order derived from any system Σ of elements of T exists uniquely. Sums and products of orders are defined correspondingly as for modules; in particular $O^2 = O$ for every order.

Every order is at the same time an R-module; an order is called finite when it is a finite module. Since sums and products arise by forming module sums and products, sums and products of finite orders are again finite orders by 1.; further, the following transitive law holds: if A is a finite R-order and B is a finite A-order, then B is also a finite R-order. For B is at the same time an R-order, hence an R-module. If $\alpha_1, \ldots, \alpha_m$ form a module basis of A with respect to R, and $\beta_1, \ldots, \beta_n$ one of B with respect to A, then the products $\alpha_i\beta_j$ plainly form a module basis of B with respect to R.

3. An element α of T is called integral with respect to R, briefly integral, if the order R_α derived from α is finite — equivalently, if the divisor chain of modules

$$U_n = (\alpha^0, \alpha, \ldots, \alpha^{n-1})$$

terminates after finitely many steps, $U_n = U_{n+1} = \cdots$.

The definition also admits a second formulation, to be used in 4.: an element α of T is called integral if there is at least one principal module C different from the zero module — i.e. C is derived from an element $\gamma \neq 0$ — such that the product of the order R_α with C is a finite module; in other words, such that the chain of modules CU_n terminates after finitely many steps.[8]

[7] The theory of integral quantities remains valid if zero divisors are admitted, provided that in their place one assumes the divisor-chain condition in R, which by §2 also makes the proof of the lemma possible. Since this form of the theory of integral quantities is not used later, I only point to it in footnotes. All definitions under discussion remain valid in the presence of zero divisors; most of them require — as is known and easy to see — even weaker assumptions. What is not retained is Dedekind's formulation: "A number is integral when it has a hull." For with zero divisors present, the quotient of two finite modules need not remain finite; hence here the order is defined directly and not as a module quotient.

[8] If zero divisors are admitted in R, then the existence of a regular principal module must be required; that is, γ must be assumed to be a non-zero-divisor, a regular element, which in rings without zero divisors is simply $\gamma \neq 0$.

Finally the definition is also identical with the usual formulation: an element α of T is called integral if it satisfies at least one equation

$$\alpha^n + r_1\alpha^{n-1} + \cdots + r_n = 0,$$

where the r's are elements of R.

These definitions are in fact identical. Since R_α agrees with the module derived from all powers of α, when R_α is finite one can always choose a module basis from these powers. If such a basis is given, for instance, by $e = \alpha^0, \alpha, \ldots, \alpha^{n-1}$, this means that the chain of the U_n terminates at U_n; conversely, $U_n = U_{n+1} = \cdots$ gives this basis. But the termination of the chain of the U_n is also identical with the existence of the equation

$$\alpha^n + r_1\alpha^{n-1} + \cdots + r_n = 0.$$

If R_α is finite, then CR_α is finite, so the chain CU_n terminates; conversely, from the finiteness of CR_α, hence from the termination of the CU_n, it follows that the U_n also terminate and hence that R_α is finite. Indeed, since C is assumed to be a principal module distinct from the zero module, $CU_n = CU_{n+1}$ also gives $U_n = U_{n+1}$.

The ring property of the integral quantities and the transitive law of integral dependence rest on the following

Lemma. Let O be a finite order in T, and let α be any element of O. Then the order R_α derived from α is finite; in other words, all elements of a finite order are integral.

The proof follows by the usual arguments. Let $\omega_1, \ldots, \omega_n$ be a module basis of O. Since O is a ring, $\alpha\omega_i$ belongs to O. Thus

$$\alpha\omega_i = r_{i1}\omega_1 + \cdots + r_{in}\omega_n,$$

and hence

$$\det(r_{ij} - e_{ij}\alpha) = 0,$$

where $e_{ij} = 0$ for $i \neq j$ and $e_{ii} = e$; this proves that α is integral. For the vanishing of the determinant one uses the hypothesis that T, and hence O, has no zero divisors.[9]

The system G of all integral quantities in T forms an order. It contains R, for the elements of R are integral, since R is a finite R-module with the identity element as basis. But G also has the ring property: for the order $R_{\alpha,\beta}$ derived from two arbitrary integral quantities α, β is equal to the product $R_\alpha R_\beta$, and is therefore finite. Hence, by the lemma, $\alpha + \beta$ and $\alpha\beta$ are integral as elements of a finite order.

Transitive law of integral dependence. If O is an order consisting of integral quantities, then every element of T integral with respect to O is also integral with respect to R. By definition α satisfies an equation

$$\alpha^m + \omega_1\alpha^{m-1} + \cdots + \omega_m = 0,$$

and is therefore also integral with respect to the order $O' = R_{\omega_1,\ldots,\omega_m}$ derived from $\omega_1, \ldots, \omega_m$, which is finite as the product $R_{\omega_1} \cdots R_{\omega_m}$. By the transitive law given in 2., the finite O'-order O'_α derived from α is therefore also a finite R-order, and α is integral by the lemma as an element of a finite order.

4. The ring R is called integrally closed in T if every element of T integral with respect to R belongs to R. According to the second formulation of the definition of integral quantities in 3., an element α of T belongs to R if and only if there is at least one principal module C different from the zero module such that CR_α is a finite module.[10]

From the concept of an integrally closed ring Dedekind (3rd ed., §172) drew two important consequences, which make it possible to use this concept for ideal theory:

Dedekind consequence I. Let R be integrally closed in T, and assume further that the divisor-chain condition for ideals (axiom I) holds in R. If β is a nonintegral element of T, and $c \neq 0$ an element of R, then there is an exponent $\rho > 0$ such that in the sequence of elements

$$c,\ c\beta, \ldots, c\beta^\nu, \ldots$$

[9] If the divisor-chain condition is assumed in R but zero divisors are admitted, then the lemma is an immediate consequence of the module theorem in §2, according to which the chain condition transfers to the modules from O. This proof too is due, for the number field, to Dedekind (4th ed., §173, II) and is the first application of chain conditions in the literature. In the consequences to be given under 4., Dedekind instead uses the more special fact that the number of residue classes modulo an ideal in an algebraic number field is finite.

[10] If zero divisors are admitted in R, C must again be assumed to be a regular principal module; likewise the element c in Consequence I must be assumed regular.

the elements $c, \dots, c\beta^\rho$ are integral, and all the remaining ones are nonintegral.

If all elements $c\beta^\nu$ were integral, then by the integral closedness of R they would belong to R. Then the chain of modules $C, CB_1, \dots, CB_\nu, \dots$ — where C denotes the module derived from c, and B_ν the one derived from $1, \beta, \dots, \beta^\nu$ — would become a chain of ideals $C_\nu = (c, c\beta, \dots, c\beta^\nu)$, which terminates by hypothesis. But then, contrary to the assumption, R_β would be finite and β integral; hence some of the elements $c\beta^\nu$ are nonintegral. Furthermore, with any one nonintegral element all subsequent ones are nonintegral: for if $c\beta^r$ is integral, hence belongs to R, then for $\rho < r$ the element $c\beta^\rho$ satisfies

$$(c\beta^\rho)^r - c^{r-\rho}(c\beta^r)^\rho = 0$$

and is therefore integral. Thus if $c\beta^{\rho+1}$ is the first nonintegral element, then — since c is integral — $\rho > 0$, and this is the required exponent.

Specializing the extension ring T to the quotient field of R — that is, to the field obtained by adjoining all pairs of elements[11] — gives the following:

Dedekind consequence II. Let R be integrally closed in its quotient field K, and let the divisor-chain condition for ideals hold in R. If β is a nonintegral element of K, then β admits a representation as a quotient of elements of R,

$$\beta = m/n,$$

such that m^2/n is also nonintegral.

Put $\beta = b/c$ with $c \neq 0$. Then in the sequence $c, c\beta = b, c\beta^2, \dots$ the first two elements are certainly integral, so $\rho > 1$, where ρ is the exponent of Consequence I. If one sets

$$m = c\beta^\rho, \qquad n = c\beta^{\rho-1},$$

then $m/n = \beta$, and $m^2/n = c\beta^{\rho+1}$ is nonintegral.

The transitive law of integral closedness has the following form: if R is any ring in T and G denotes the system of all quantities in T integral with respect to R, then G also consists of all quantities of T integral with respect to G. For every quantity integral with respect to G is also integral with respect to R, by the transitive law in 3., and hence belongs to G.

§2. Chain Conditions in Finite Module Domains

The module theorem by which the chain conditions are transferred occurs, in the application given here, only for modules of an extension ring. Since the proof is the same in the case of a general module domain, such a domain will be taken as the starting point.

A system M of elements $\alpha, \beta, \dots$ is called a module domain with respect to a ring R if two operations are given in M, each leading uniquely to elements of M: an additive combination of the elements and a multiplication of the elements by elements of R; if M is an Abelian group with respect to addition, while multiplication satisfies the associative law; and if the distributive law holds in both forms.[12]

For a module domain, all module definitions given in §1, 1. that do not refer to multiplication plainly remain valid. In particular, M is called a finite module domain if there are finitely many elements $\xi_1, \dots, \xi_k$ of M such that M equals the module derived from $\xi_1, \dots, \xi_k$, hence equals the system of all linear forms $r_1\xi_1 + \dots + r_k\xi_k$ when R has an identity element; otherwise additional terms $n_i\xi_i$ occur.

Module theorem. Let M be a finite module domain with respect to a commutative ring R with identity element, and suppose that in R the divisor- respectively multiple-chain condition for ideals holds. Then in M the divisor- respectively multiple-chain condition for modules holds.

The proof rests on assigning uniquely to every module in M a system of finitely many ideals from R in such a way that to a proper divisor respectively proper multiple of the module there corresponds each time at least one proper divisor respectively multiple ideal.

An element of M is said to have length i if it can be written in at least one way as a linear form in $\xi_1, \dots, \xi_i$, while it has no representation as a linear form in $\xi_1, \dots, \xi_{i-1}$. To an arbitrary module W in M assign now k ideals $\mathfrak{a}_1, \dots, \mathfrak{a}_k$ from R: let W_i mean the module of all elements of W of length $\leq i$, and let $\mathfrak{a}_i$ denote the system of all coefficients of ξ_i in W_i. First one obtains:

[11] In Steinitz's familiar formulation, J. f. M. 137. If zero divisors are admitted in R, then the quotient ring must replace the quotient field, by adjoining all quotient pairs whose denominator is a regular element of R. Here Dedekind consequence II is no longer valid, for n will not in general be a regular element.

[12] Compare *Ideal Theory*, §9.

If B is a proper divisor of W, and if $\mathfrak{b}_1, \dots, \mathfrak{b}_k$ are the ideals assigned to B, then among the $\mathfrak{b}_i$ there is at least one proper divisor of the corresponding $\mathfrak{a}_i$. For from $B \equiv 0 \pmod{W}$ it follows that $B_i \equiv 0 \pmod{W_i}$, and hence $\mathfrak{b}_i \equiv 0 \pmod{\mathfrak{a}_i}$ for $i = 1, 2, \dots, k$. By hypothesis there must be elements in B not contained in W, and hence such elements of smallest length ℓ. Then $\mathfrak{b}_\ell$ is a proper divisor of $\mathfrak{a}_\ell$; indeed $\mathfrak{b}_\ell \equiv 0 \pmod{\mathfrak{a}_\ell}$ when

$$\beta = b_1\xi_1 + \cdots + b_\ell\xi_\ell$$

is such an element of smallest length in B. If $\mathfrak{b}_\ell = \mathfrak{a}_\ell$, there exists an element

$$\alpha = a_\ell\xi_\ell + \cdots + a_1\xi_1 \equiv 0 \pmod{W},$$

and therefore an element $\beta - \alpha \in B$, $\beta - \alpha \notin W$, of smaller length than ℓ.

Now let a divisor or multiple chain

$$W^{(1)}, W^{(2)}, \dots, W^{(\nu)}, \dots$$

be given; suppose the corresponding chain condition holds in R. If $\mathfrak{a}_{1\nu}, \dots, \mathfrak{a}_{k\nu}$ are the ideals assigned to $W^{(\nu)}$, then the sequences

$$\mathfrak{a}_{i1}, \mathfrak{a}_{i2}, \dots$$

are divisor respectively multiple chains for $i = 1, \dots, k$. Hence by hypothesis there is an index ν_0 such that the ideal $\mathfrak{a}_{i\nu_0}$ is equal to all subsequent ideals for every i. But then, by what was just proved, the module $W^{(\nu_0)}$ is equal to all subsequent modules; thus the chain conditions are transferred.

This immediately gives the following:

Consequence of the module theorem. If in R the multiple-chain condition holds modulo every ideal different from the zero ideal, then in M the multiple-chain condition holds modulo every module C whose assigned ideals $\mathfrak{c}_1, \dots, \mathfrak{c}_k$ are all different from the zero ideal. Under these hypotheses the multiple chains $\mathfrak{a}_{i1}, \dots, \mathfrak{a}_{i\nu}, \dots$ terminate after finitely many steps.

Additional remark. If R is a ring without identity element, then the divisor-chain condition and the multiple-chain condition modulo ideals different from the zero ideal are still transferred. Namely, instead of M one considers the system M' of all elements of the form

$$a_1\xi_1 + \cdots + a_k\xi_k + n_1\xi_1 + \cdots + n_k\xi_k,$$

which again passes into M when the additional integral multiples $n_i\xi_i$ are replaced by elements of M. To this M' one can assign, correspondingly as above, $2k$ ideals, where the $\mathfrak{a}_i$ are ideals from R and the $\mathfrak{n}_i$ are ideals of the rational integers. Since for the $\mathfrak{n}_i$ both the divisor-chain condition and the multiple-chain condition modulo every nonzero ideal are satisfied, the proof for the divisor-chain condition remains valid for M' and hence for M; for the multiple-chain condition, however, one must require that the $2k$ ideals assigned to C respectively C' are all different from the zero ideal.

§3. Passage to finite extension fields

For the classification of number fields and function fields it remains to show how the axioms I through V formulated in the introduction are transferred on passing to finite extension fields.

1. *Suppose that in $\mathfrak{R}$ all axioms I through V are satisfied except axiom II – the multiple-chain condition. Let $\mathfrak{K}$ denote the quotient field of $\mathfrak{R}$, and let $\mathfrak{L}$ be a finite extension field of the first kind over $\mathfrak{K}$.*[14] *Then in the system $\mathfrak{S}$ of all elements of $\mathfrak{L}$ integral with respect to $\mathfrak{R}$ the same axioms are satisfied, and in every order from $\mathfrak{S}$ all these axioms are still satisfied with the exception of integral closedness.*

By §1, 3, $\mathfrak{S}$ forms a ring for which axioms III and IV – existence of the identity element and absence of zero divisors – are satisfied. By the conclusion of §1, 4, $\mathfrak{S}$ is integrally closed in $\mathfrak{L}$; at the same time $\mathfrak{L}$ becomes the quotient field of $\mathfrak{S}$, since every element of $\mathfrak{L}$ is carried into an element of $\mathfrak{S}$ by multiplication with a suitable element of $\mathfrak{R}$. Thus axiom V is satisfied.

The proof of I follows by familiar arguments. As an extension of the first kind, $\mathfrak{L}$ is obtained by adjoining to $\mathfrak{K}$ a single element α, which by the preceding may be assumed integral with respect to $\mathfrak{R}$. Besides α, the conjugate quantities $\alpha', \alpha'', \dots$ contained in the Galois field of $\mathfrak{L}$ over $\mathfrak{K}$ are also integral; hence so is their

[14] Following Steinitz, an algebraic extension field is called "of the first kind" if every element is a zero of a prime function which, in a suitable extension field, splits into pairwise distinct linear factors.

product of differences, and so is the square D of this product of differences. Since $\alpha, \alpha', \ldots$ are all distinct, D is a nonzero quantity of $\mathfrak{K}$, hence, by the integral closedness of $\mathfrak{R}$ in $\mathfrak{K}$, an element of $\mathfrak{R}$. By the usual argument, because $\mathfrak{R}$ is integrally closed, $\mathfrak{S}$ is contained in the finite $\mathfrak{R}$-module domain M generated by the elements $\xi_i = \alpha^i/D$; for this one need only pass, in the representation of an element of $\mathfrak{S}$ by the ξ_i, to the conjugate elements and solve the resulting system of linear equations for the coefficients of the representation. By the module theorem of §2, the divisor-chain condition holds for all $\mathfrak{R}$-modules from M, hence in particular for all ideals of $\mathfrak{S}$. The divisor-chain condition likewise holds for the ideals of any order in $\mathfrak{S}$, since here too one is dealing with $\mathfrak{R}$-modules in M; for every order, moreover, axioms III and IV are satisfied.

2. *If all axioms I through V are satisfied in $\mathfrak{R}$, the same is true in $\mathfrak{S}$. In every order from $\mathfrak{S}$ all axioms still hold except integral closedness.*

In view of 1, it remains only to prove axiom II. By the corollary to the module theorem in §2 it is enough to prove the following. If a nonzero ideal of $\mathfrak{S}$ is regarded as an $\mathfrak{R}$-module $\mathfrak{C}$ in M, then the ideals $\mathfrak{c}_i$ of $\mathfrak{R}$ associated with $\mathfrak{C}$ are all different from the zero ideal. Now $\mathfrak{C}$ has the same finite linear rank over $\mathfrak{K}$ as M; for along with any element γ, $\mathfrak{C}$ contains also $\gamma\alpha^i$. Thus for each ξ_i there is an element $c_i \neq 0$ of $\mathfrak{R}$ such that $c_i\xi_i$ belongs to $\mathfrak{C}$; by the uniqueness of the representation by the ξ_i – where the index is to run only through the values less than the degree of the field – c_i belongs to $\mathfrak{c}_i$, and therefore $\mathfrak{c}_i$ is not the zero ideal. This argument remains valid for all orders of the same rank as M; for an order of smaller rank one passes to its quotient field, which as an intermediate field between $\mathfrak{K}$ and $\mathfrak{L}$ is also of the first kind, and whose degree over $\mathfrak{K}$ agrees with the linear rank of the order. The same argument is therefore preserved. Finally, it should be noted that an order in $\mathfrak{S}$ different from $\mathfrak{S}$ is never integrally closed: since every order also contains $\mathfrak{R}$, integral closedness would force it to contain $\mathfrak{S}$.

For the subordination of number fields and function fields, it is therefore enough to verify axioms I through V for the basic domains: the rational integers, polynomials in one indeterminate, and the functional domain of polynomials in several indeterminates.

3. *For the ring $\mathfrak{R}$ of rational integers, respectively of polynomials in one indeterminate with coefficients in a field, axioms I through V are satisfied.* Here I and II follow either from the fact that, modulo any nonzero number or any such polynomial in one indeterminate, there are only finitely many, respectively finitely many linearly independent, residue classes; or from the fact that every ideal is principal. For from this follows the unique decomposition of ideals as a product of powers of prime ideals, and hence a nonzero ideal is divisible by only finitely many ideals. Axioms III and IV are plainly satisfied, the latter as a consequence of the definition of equality. Integral closedness V follows from the fact that, by unique factorization of elements of $\mathfrak{R}$ into irreducibles up to units – divisors of one – every element of the quotient field not contained in $\mathfrak{R}$ can be written as a quotient of two relatively prime elements of $\mathfrak{R}$; hence no power of the numerator is divisible by the denominator. Such an element can therefore satisfy no equation characterizing it as integral with respect to $\mathfrak{R}$.

4. *Let $\mathfrak{R}$ now denote the ring of all polynomials in several indeterminates with coefficients in a field, or with rational integer coefficients.* Then axioms III, IV, V are satisfied as in 3, for here too the factorization of elements into irreducibles is unique up to units. The divisor-chain condition I is an immediate consequence of Hilbert's theorem on the existence of an ideal basis, whereas the multiple-chain condition II does not hold here.

However, by passing to the *functional domain*, which amounts to restricting to ideals of highest dimension, one can obtain the validity of axiom II as well; the validity of I can then be proved as in 3, without using Hilbert's theorem. Namely, adjoin to $\mathfrak{R}$ an indeterminate u and consider the ring $\mathfrak{R}^*$ of all polynomials in u with coefficients in $\mathfrak{R}$, equality being defined coefficientwise. As usual, call a polynomial in $\mathfrak{R}^*$ primitive (with respect to $\mathfrak{R}$) if the greatest common divisor of its coefficients – divisor in the sense of a polynomial in $\mathfrak{R}$, not an ideal divisor – is a unit of $\mathfrak{R}$. Then every polynomial in $\mathfrak{R}^*$ is the product of an element of $\mathfrak{R}$ and a primitive polynomial in $\mathfrak{R}^*$, and the product of primitive polynomials in $\mathfrak{R}^*$ is again primitive.

The totality of elements of the quotient field of $\mathfrak{R}^*$ whose denominators are primitive polynomials in $\mathfrak{R}^*$ therefore forms a ring, the functional domain $\mathfrak{F}$ of $\mathfrak{R}$. In this functional domain $\mathfrak{F}$, the units are exactly those elements whose numerator and denominator are primitive polynomials in $\mathfrak{R}^*$; every element of $\mathfrak{F}$ is therefore uniquely representable as the product of an element of $\mathfrak{R}$ and a unit of $\mathfrak{F}$. Hence the factorization theorem for the elements of $\mathfrak{R}$ transfers to the elements of $\mathfrak{F}$; for $\mathfrak{F}$, therefore, in addition to axioms III and IV, axiom V is also satisfied as in 3.

Moreover, in $\mathfrak{F}$ every ideal is principal. For together with any element of $\mathfrak{F}$ an ideal contains also the element of $\mathfrak{R}$ obtained from it by multiplying by a unit; and together with any two elements of $\mathfrak{R}$ it contains their greatest common divisor. Indeed, with $f(x) = t(x)\overline{f}(x)$ and $g(x) = t(x)\overline{g}(x)$ it also contains $t(x)(\overline{f}(x)+u\overline{g}(x))$, and hence $t(x)$, if $\overline{f}$ and $\overline{g}$ are relatively prime and their linear combination is consequently a unit. Thus, starting from an arbitrary element $f(x)$ of the ideal, if there is in the ideal an element $g(x)$ not divisible by it, then the greatest common divisor $t(x)$ also belongs to the ideal and is a proper divisor of $f(x)$. Repeating this finitely many times leads to a basis element of the ideal, and so the ideal is recognized as principal. Thus in $\mathfrak{F}$ the unique decomposition of ideals as products of powers of prime ideals holds, corresponding to the factorization of the elements of $\mathfrak{R}$; as in 3, it follows that axioms I and II are satisfied. The extension fields of the quotient field of $\mathfrak{F}$ that come into consideration here are only those that can be generated by adjoining an element of $\mathfrak{S}$.[15] Taking 1 and 2 into account, this establishes the validity of axioms I through V for number fields and function fields.

§4. Isomorphism theorems. Direct sums

In what follows only the ring property, respectively the module property, is assumed; no further axiom is presupposed.

1. If M and $\overline{M}$ are module domains with respect to $\mathfrak{R}$ (§2), M is called homomorphic to $\overline{M}$ (more precisely, module-homomorphic), $M \sim \overline{M}$, if to every element of M there corresponds one and only one element of $\overline{M}$, in such a way that $\overline{M}$ is exhausted;[16] and if, under this correspondence, difference and multiplication by the same element of $\mathfrak{R}$ correspond. Thus from $\beta \sim \overline{\beta}$ and $\gamma \sim \overline{\gamma}$ it always follows that $(\beta - \gamma) \sim (\overline{\beta} - \overline{\gamma})$ and $r\beta \sim r\overline{\beta}$.

An $\mathfrak{R}$-module $\mathfrak{B}$ in M therefore corresponds homomorphically to an $\mathfrak{R}$-module $\overline{\mathfrak{B}}$ in $\overline{M}$; and the totality $\mathfrak{C}^*$ of elements in M corresponding to elements of an $\mathfrak{R}$-module $\overline{\mathfrak{C}}$ in $\overline{M}$ forms an $\mathfrak{R}$-module in M uniquely determined by $\overline{\mathfrak{C}}$, the module $\mathfrak{C}^*$ associated with $\overline{\mathfrak{C}}$, with $\mathfrak{C}^* \sim \overline{\mathfrak{C}}$. If in particular $\mathfrak{A}$ is the module in M associated with the zero element of $\overline{M}$, then $\mathfrak{C}^*$ is a divisor of $\mathfrak{A}$ and splits into classes of elements of M congruent modulo $\mathfrak{A}$, in such a way that these classes correspond one-to-one to the elements of $\overline{\mathfrak{C}}$. If one passes from $\mathfrak{B}$ in M to $\overline{\mathfrak{B}}$ in $\overline{M}$ and then back to $\mathfrak{B}^*$, one obtains $\mathfrak{B}^* = (\mathfrak{B}, \mathfrak{A})$, the greatest common divisor of $\mathfrak{B}$ and $\mathfrak{A}$; thus $\mathfrak{B}^* = \mathfrak{B}$ if $\mathfrak{B}$ is a divisor of $\mathfrak{A}$.

If the correspondence between the elements of M and $\overline{M}$ is reversible one-to-one, the domains are called isomorphic (more precisely, module-isomorphic), $M \cong \overline{M}$.

If $\mathfrak{A}$ is an arbitrary $\mathfrak{R}$-module in M, then a module domain $\overline{M}$ homomorphic to M arises – the residue-class module $M \mid \mathfrak{A}$ – by taking congruence modulo $\mathfrak{A}$ as the new equality relation. To every element of M there are associated all and only the elements equal to it in $\overline{M}$. Passing in $\overline{M}$ from the equality definition to identity means collecting all equal elements in $\overline{M}$ into one class – a residue class – and taking these residue classes as the new elements of $\overline{M}$.

Every homomorphism is produced by passage to a residue-class module; for if $M \sim \overline{M}$ and if $\mathfrak{A}$ is the module associated with the zero element of $\overline{M}$, then, as shown above, $\overline{M}$ is isomorphic to the residue-class module $M \mid \mathfrak{A}$.

2. **First isomorphism theorem.** *Let $\overline{M}$ be the residue-class module $M \mid \mathfrak{A}$, and let $\mathfrak{C}$ be a divisor of $\mathfrak{A}$. Then there is an isomorphism $\overline{M} \mid \overline{\mathfrak{C}} \cong M \mid \mathfrak{C}$.* Indeed, congruence modulo $\mathfrak{A}$ is at the same time congruence modulo $\mathfrak{C}$; elements equal modulo $\mathfrak{A}$ remain equal modulo $\mathfrak{C}$. Thus one may form the residue-class module $M \mid \mathfrak{C}$ – that is, equality modulo $\mathfrak{C}$ – by first identifying elements modulo $\mathfrak{A}$, passing to $\overline{M}$, and then collecting in $\overline{M}$ those elements that are equal modulo $\mathfrak{C}$; this is the same as identifying modulo $\overline{\mathfrak{C}}$ in $\overline{M}$, i.e. forming $\overline{M} \mid \overline{\mathfrak{C}}$.

Second isomorphism theorem. *If $\mathfrak{B}$ and $\mathfrak{A}$ are modules in M, then* $(\mathfrak{B}, \mathfrak{A}) \mid \mathfrak{A} \cong \mathfrak{B} \mid [\mathfrak{B}, \mathfrak{A}]$. For by 1, $\mathfrak{B}$ becomes homomorphic to $\overline{\mathfrak{B}}$ when $(\mathfrak{B}, \mathfrak{A}) \mid \mathfrak{A}$ is put equal to $\overline{\mathfrak{B}}$; and since precisely the elements of $[\mathfrak{B}, \mathfrak{A}]$ correspond to the zero element in $\overline{\mathfrak{B}}$, the asserted isomorphism again follows from 1.

3. If $\mathfrak{R}$ and $\overline{\mathfrak{R}}$ are (commutative) rings, $\mathfrak{R}$ is called homomorphic to $\overline{\mathfrak{R}}$ (more precisely, ring-homomorphic), $\mathfrak{R} \sim \overline{\mathfrak{R}}$, if to each element of $\mathfrak{R}$ there corresponds one and only one element of $\overline{\mathfrak{R}}$ in such a way that $\overline{\mathfrak{R}}$ is exhausted, and if differences and products correspond under this assignment. If the correspondence of elements is reversible one-to-one, the rings are called isomorphic (more precisely, ring-isomorphic), $\mathfrak{R} \cong \overline{\mathfrak{R}}$.

All the arguments in 1 and 2 remain valid when the modules are replaced by ideals in $\mathfrak{R}$,[17] and when module-homomorphism and module-isomorphism are replaced by ring-homomorphism and ring-isomorphism.

[15] The system of quantities integral with respect to $\mathfrak{F}$ in these extension fields is also obtained by passing from $\mathfrak{S}$ to a corresponding functional domain, just as one passes from $\mathfrak{R}$ to $\mathfrak{F}$. Compare J. König, *Algebraische Größen*, Leipzig 1903, pp. 468/69.

[16] In the sense of the definition of equality prevailing in $\overline{M}$; compare note 6.

[17] For noncommutative rings, two-sided ideals must be used.

In particular, if $\mathfrak{c}$ is a divisor of $\mathfrak{a}$, the residue-class ring $\mathfrak{c} \mid \mathfrak{a}$ is obtained by introducing congruence modulo $\mathfrak{a}$ as the new equality relation; here one must note that products of residue classes are now also defined.

Thus $\mathfrak{c}$ is homomorphic to $\mathfrak{c} \mid \mathfrak{a}$; every homomorphism is produced in this way, and the isomorphism theorems hold as in 2:

First isomorphism theorem. If $\overline{\mathfrak{R}}$ denotes the residue-class ring $\mathfrak{R} \mid \mathfrak{a}$, and $\mathfrak{c}$ is a divisor of $\mathfrak{a}$, then $\overline{\mathfrak{R}} \mid \bar{\mathfrak{c}} \cong \mathfrak{R} \mid \mathfrak{c}$ in the sense of ring isomorphism.

Second isomorphism theorem. If $\mathfrak{b}$ and $\mathfrak{a}$ are ideals of $\mathfrak{R}$, then the ring-isomorphism $(\mathfrak{b}, \mathfrak{a}) \mid \mathfrak{a} \cong \mathfrak{b} \mid [\mathfrak{b}, \mathfrak{a}]$ holds.[18]

4. I now collect the calculation rules for coprime ideals[19] from which the theorems on direct sums in 5 follow. In the commutative ring $\mathfrak{R}$, assume the existence of a multiplicative identity element e. Thus $\mathfrak{a} = \mathfrak{o}\mathfrak{a}$ for every ideal of $\mathfrak{R}$, where $\mathfrak{o}$ denotes the unit ideal. Two ideals $\mathfrak{a}, \mathfrak{b}$ are called coprime if their greatest common divisor $(\mathfrak{a}, \mathfrak{b})$ is $\mathfrak{o}$.

4α. If $(\mathfrak{a}, \mathfrak{b}) = \mathfrak{o}$*, then* $(\mathfrak{a}, \mathfrak{b}\mathfrak{c}) = (\mathfrak{a}, \mathfrak{c})$. For $(\mathfrak{a}, \mathfrak{c}) = (\mathfrak{a}, \mathfrak{b})(\mathfrak{a}, \mathfrak{c}) = (\mathfrak{a}^2, \mathfrak{a}\mathfrak{b}, \mathfrak{a}\mathfrak{c}, \mathfrak{b}\mathfrak{c})$ is divisible by $(\mathfrak{a}, \mathfrak{b}\mathfrak{c})$ and conversely. Hence:

4β. From $\mathfrak{c}\mathfrak{b} \equiv 0 \pmod{\mathfrak{a}}$ *and* $(\mathfrak{a}, \mathfrak{b}) = \mathfrak{o}$ *it follows that* $\mathfrak{c} \equiv 0 \pmod{\mathfrak{a}}$. Indeed, $\mathfrak{c}\mathfrak{b} \equiv 0 \pmod{\mathfrak{a}}$ is equivalent to $(\mathfrak{a}, \mathfrak{b}\mathfrak{c}) = \mathfrak{a}$; because $(\mathfrak{a}, \mathfrak{b}) = \mathfrak{o}$, this also gives $(\mathfrak{a}, \mathfrak{c}) = \mathfrak{a}$, or $\mathfrak{c} \equiv 0 \pmod{\mathfrak{a}}$. Furthermore:

4γ. If each of the ideals $\mathfrak{a}_1, \ldots, \mathfrak{a}_r$ *is coprime to each of the ideals* $\mathfrak{b}_1, \ldots, \mathfrak{b}_s$*, then the product of the* $\mathfrak{a}_i$ *is coprime to the product of the* $\mathfrak{b}_i$. For from $(\mathfrak{a}, \mathfrak{b}) = \mathfrak{o}$ and $(\mathfrak{a}, \mathfrak{c}) = \mathfrak{o}$ one obtains $(\mathfrak{a}, \mathfrak{b}\mathfrak{c}) = \mathfrak{o}$, and finite repetition gives the assertion.

From 4α and 4γ follows:

4δ. If the ideals $\mathfrak{b}_1, \ldots, \mathfrak{b}_r$ *are pairwise coprime, i.e.* $(\mathfrak{b}_i, \mathfrak{b}_k) = \mathfrak{o}$ *for* $i \neq k$*, then their least common multiple equals their product.* Let $(\mathfrak{a}, \mathfrak{b}) = \mathfrak{o}$ and $\mathfrak{v} = [\mathfrak{a}, \mathfrak{b}]$. Then $\mathfrak{v} = \mathfrak{o}\mathfrak{v} = (\mathfrak{a}\mathfrak{v}, \mathfrak{b}\mathfrak{v}) = 0(\mathfrak{a}, \mathfrak{b})$; since $\mathfrak{a}\mathfrak{b} \equiv 0 \pmod{\mathfrak{v}}$, it follows that $\mathfrak{v} = \mathfrak{a}\mathfrak{b}$. Finite repetition, using 4γ, gives the claim.

4ε. If the ideals $\mathfrak{b}_1, \ldots, \mathfrak{b}_r$ *are pairwise coprime and* $\mathfrak{a}_i = [\mathfrak{b}_1, \ldots, \mathfrak{b}_{i-1}, \mathfrak{b}_{i+1}, \ldots, \mathfrak{b}_r] = \mathfrak{b}_1 \cdots \mathfrak{b}_{i-1}\mathfrak{b}_{i+1} \cdots \mathfrak{b}_r$*, then* $(\mathfrak{a}_1, \ldots, \mathfrak{a}_{i-1}, \mathfrak{a}_{i+1}, \ldots, \mathfrak{a}_r) = \mathfrak{b}_i$*, and hence* $(\mathfrak{a}_1, \ldots, \mathfrak{a}_r) = \mathfrak{o}$. For by the distributive law $(\mathfrak{a}_1, \mathfrak{a}_2) = \mathfrak{b}_3 \cdots \mathfrak{b}_r(\mathfrak{b}_2, \mathfrak{b}_1) = \mathfrak{b}_3 \cdots \mathfrak{b}_r$; hence $(\mathfrak{a}_1, \mathfrak{a}_2, \mathfrak{a}_3) = \mathfrak{b}_4 \cdots \mathfrak{b}_r(\mathfrak{b}_3, \mathfrak{b}_1\mathfrak{b}_2) = \mathfrak{b}_4 \cdots \mathfrak{b}_r$, and finite repetition, after a suitable numbering, proves the assertion.

The ideal $\mathfrak{a}_i$ is called the complement of $\mathfrak{b}_i$ in the representation $\mathfrak{m} = [\mathfrak{b}_1, \ldots, \mathfrak{b}_r]$.

5. Again let the commutative ring $\mathfrak{R}$ have a multiplicative identity element. The ring $\mathfrak{R}$ is called the direct sum of the ideals $\mathfrak{a}_1, \ldots, \mathfrak{a}_r$, in symbols $\mathfrak{R} = \mathfrak{a}_1 + \mathfrak{a}_2 + \cdots + \mathfrak{a}_r$, if each element c of $\mathfrak{R}$ can be represented in one and only one way in the form $c = a_1 + a_2 + \cdots + a_r$, with a_i an element of $\mathfrak{a}_i$.

If the zero ideal is the least common multiple of the pairwise coprime ideals $\mathfrak{b}_1, \ldots, \mathfrak{b}_r$*, and if* $\mathfrak{a}_i$ *is the complement of* $\mathfrak{b}_i$*, then* $\mathfrak{R}$ *is the direct sum of the ideals* $\mathfrak{a}_1, \ldots, \mathfrak{a}_r$. Since $\mathfrak{o} = (\mathfrak{a}_1, \ldots, \mathfrak{a}_r)$, every element of $\mathfrak{R}$ admits at least one additive representation by the $\mathfrak{a}_i$. Since $[\mathfrak{a}_i, \mathfrak{b}_i] = (0)$, this representation is unique: from $0 = a_1 + a_2 + \cdots + a_r = a_i + b_i$ one obtains $a_i = 0$. By the second isomorphism theorem the $\mathfrak{a}_i$ are isomorphic to the residue-class rings $\mathfrak{R} \mid \mathfrak{b}_i$. The orthogonality relations $\mathfrak{a}_i\mathfrak{a}_k = (0)$ for $i \neq k$, and $\mathfrak{a}_i^2 = \mathfrak{a}_i$, hold; the latter because $\mathfrak{a}_i = \mathfrak{o}\mathfrak{a}_i$.

Conversely, if there is a representation of $\mathfrak{R}$ *as a direct sum,* $\mathfrak{R} = \mathfrak{a}_1 + \mathfrak{a}_2 + \cdots + \mathfrak{a}_r$*, and if one sets* $\mathfrak{b}_i = (\mathfrak{a}_1, \ldots, \mathfrak{a}_{i-1}, \mathfrak{a}_{i+1}, \ldots, \mathfrak{a}_r) = \mathfrak{a}_1 + \cdots + \mathfrak{a}_{i-1} + \mathfrak{a}_{i+1} + \cdots + \mathfrak{a}_r$*, then the* $\mathfrak{b}_i$ *are pairwise coprime and their least common multiple is the zero ideal.* For from $\mathfrak{o} = (\mathfrak{a}_i, \mathfrak{b}_i)$ follows $\mathfrak{o} = (\mathfrak{b}_k, \mathfrak{b}_i)$ for $k \neq i$, since $\mathfrak{b}_k$ is a divisor of $\mathfrak{a}_i$. If, moreover, c is divisible by all $\mathfrak{b}_i$, then

$$c = a_2^{(1)} + \cdots + a_r^{(1)} = a_1^{(2)} + a_3^{(2)} + \cdots + a_r^{(2)} = \cdots = a_1^{(r)} + \cdots + a_{r-1}^{(r)},$$

where in each case $a_i^{(\lambda)} \equiv 0 \pmod{\mathfrak{a}_i}$. By the uniqueness of the representation, since in each representation one component is zero, it follows that $0 = a_1^{(\lambda)}, \ldots, 0 = a_r^{(\lambda)}$ for every λ, and hence $c = 0$.[20]

[18] These isomorphism theorems, familiar from group theory, occur for modules in a somewhat more special form first in Dedekind (compare 3rd ed., p. 484). The above form for ideals occurs in Masazo Sono, *On Congruences. I, II, III, IV*, Memoirs of the College of Science, Kyoto Imperial University, 2 (1917), 3 (1918, 1919), compare 2, p. 215. In each case only the second isomorphism theorem is stated explicitly.

[19] In the form going back to Dedekind (4th ed., §178, III to VIII). The calculation with the identity element that appears throughout more recent accounts is much more cumbersome.

[20] The theorems given under 5 are special cases of a general theorem on the connection between direct sum and direct intersection. Compare H. Prüfer, Theorie der Abelschen Gruppen I, Math. Zeitschr. 20 (1924), pp. 165–187, §6. The arguments given there remain valid also for such a general formulation of the concept of group that modules and ideals fall under it as special cases.

§5. Prime ideals and primary ideals

Let $\mathfrak{R}$ again be a commutative ring for which no further axiom, not even the existence of an identity element, is assumed. We collect the basic facts about prime ideals and primary ideals.[21]

1. An ideal $\mathfrak{p}$ of $\mathfrak{R}$ is called a *weak prime ideal* if the residue-class ring $\mathfrak{R} \mid \mathfrak{p}$ is a ring without zero divisors; that is, if from $\mathfrak{a} \not\equiv 0 \pmod{\mathfrak{p}}$ and $\mathfrak{b} \not\equiv 0 \pmod{\mathfrak{p}}$ it always follows that $\mathfrak{ab} \not\equiv 0 \pmod{\mathfrak{p}}$. An ideal $\mathfrak{p}^*$ of $\mathfrak{R}$ is called a *strong prime ideal* if the residue-class ring $\mathfrak{R} \mid \mathfrak{p}^*$ is a ring without zero-divisor ideals; that is, if from $\mathfrak{a} \not\equiv 0 \pmod{\mathfrak{p}^*}$ and $\mathfrak{b} \not\equiv 0 \pmod{\mathfrak{p}^*}$ it always follows that $\mathfrak{ab} \not\equiv 0 \pmod{\mathfrak{p}^*}$.

Every strong prime ideal is at the same time a weak prime ideal, as the specialization of $\mathfrak{a}, \mathfrak{b}$ to principal ideals shows. Conversely, *every weak prime ideal is also a strong prime ideal*; one can therefore simply speak of prime ideals. For let $\mathfrak{p}$ be a weak prime ideal; let $\mathfrak{a} \not\equiv 0 \pmod{\mathfrak{p}}$ and $\mathfrak{b} \not\equiv 0 \pmod{\mathfrak{p}}$; choose elements a and b in $\mathfrak{a}$ and $\mathfrak{b}$ such that $a \not\equiv 0 \pmod{\mathfrak{p}}$ and $b \not\equiv 0 \pmod{\mathfrak{p}}$. Then $ab \not\equiv 0 \pmod{\mathfrak{p}}$, hence $\mathfrak{ab} \not\equiv 0 \pmod{\mathfrak{p}}$, so $\mathfrak{p}$ is also a strong prime ideal.

2. An ideal $\mathfrak{q}$ of $\mathfrak{R}$ is called a *weak primary ideal* if in the residue-class ring $\mathfrak{R} \mid \mathfrak{q}$ some power of every zero divisor vanishes; equivalently, if from $a \not\equiv 0 \pmod{\mathfrak{q}}$ and $b^x \not\equiv 0 \pmod{\mathfrak{q}}$ for every x it always follows that $ab \not\equiv 0 \pmod{\mathfrak{q}}$. The system $\mathfrak{p}$ of all elements of $\mathfrak{R}$ that become zero divisors in $\mathfrak{R} \mid \mathfrak{q}$ forms an ideal, indeed a prime ideal; it is a divisor of $\mathfrak{q}$ and is called the associated prime ideal. For along with a, also ra is a zero divisor; along with a and b, so is $a - b$, since with a^x and b^λ the element $(a-b)^{x+\lambda}$ is always divisible by $\mathfrak{q}$. If furthermore a is not a zero divisor, then by definition no power of a is a zero divisor; from $a \not\equiv 0 \pmod{\mathfrak{p}}$ and $b \not\equiv 0 \pmod{\mathfrak{p}}$ follows $a^x b^\lambda = (ab)^x \not\equiv 0 \pmod{\mathfrak{q}}$, hence $ab \not\equiv 0 \pmod{\mathfrak{p}}$.

An ideal $\mathfrak{q}^*$ of $\mathfrak{R}$ is called a *strong primary ideal* if in the residue-class ring $\mathfrak{R} \mid \mathfrak{q}^*$ some power of every zero-divisor ideal vanishes; that is, if from $\mathfrak{a} \not\equiv 0 \pmod{\mathfrak{q}^*}$ and $\mathfrak{b}^x \not\equiv 0 \pmod{\mathfrak{q}^*}$ for every x it always follows that $\mathfrak{ab} \not\equiv 0 \pmod{\mathfrak{q}^*}$. As for weak primary ideals one shows: the greatest common divisor of all ideals of $\mathfrak{R}$ that become zero-divisor ideals in $\mathfrak{R} \mid \mathfrak{q}^*$ forms a prime ideal, a divisor of $\mathfrak{q}^*$, the associated prime ideal. Conversely, all primary ideals with the same associated prime ideal $\mathfrak{p}$ are said to belong to $\mathfrak{p}$.

Every strong primary ideal is at the same time a weak primary ideal, as the specialization of $\mathfrak{a}, \mathfrak{b}$ to principal ideals shows. In general, however, the converse is false.[22] *If, however, axiom I, the divisor-chain condition, is assumed in $\mathfrak{R}$, then every weak primary ideal is also a strong primary ideal;* one can therefore simply speak of primary ideals. Let $\mathfrak{q}$ be a weak primary ideal; suppose $\mathfrak{a} \not\equiv 0 \pmod{\mathfrak{q}}$ and $\mathfrak{b}^x \not\equiv 0 \pmod{\mathfrak{q}}$ for every x. Then there exist elements a of $\mathfrak{a}$ and b of $\mathfrak{b}$ such that $a \not\equiv 0 \pmod{\mathfrak{q}}$ and $b^x \not\equiv 0 \pmod{\mathfrak{q}}$ for every x; hence $ab \not\equiv 0 \pmod{\mathfrak{q}}$ and therefore $\mathfrak{ab} \not\equiv 0 \pmod{\mathfrak{q}}$. For if to every b in $\mathfrak{b}$ there were an exponent λ such that $b^\lambda \equiv 0 \pmod{\mathfrak{q}}$, then $\mathfrak{b}^{\lambda_1 + \cdots + \lambda_r} \equiv 0 \pmod{\mathfrak{q}}$, with $\lambda_1, \ldots, \lambda_r$ the exponents of the finitely many basis elements.[23] The same argument also shows that there is a smallest exponent ϱ such that $\mathfrak{p}^\varrho \equiv 0 \pmod{\mathfrak{q}}$, where $\mathfrak{p}$ is the associated prime ideal; ϱ is called the exponent of $\mathfrak{q}$.

3. In what follows the divisor-chain condition is again *not* assumed. The least common multiple $[\mathfrak{a}_1, \ldots, \mathfrak{a}_r]$ of finitely many ideals is called a *shortest representation* if no $\mathfrak{a}_i$ is contained in the least common multiple of the remaining ideals, i.e. if no $\mathfrak{a}_i$ can be omitted.

The least common multiple of finitely many weak, respectively strong, primary ideals belonging to the same prime ideal $\mathfrak{p}$ is again a weak primary ideal with $\mathfrak{p}$ as associated prime ideal. A shortest representation by finitely many weak, respectively strong, primary ideals belonging to distinct prime ideals is not a weak, respectively strong, primary ideal. Let $\mathfrak{f} = [\mathfrak{q}_1, \ldots, \mathfrak{q}_r]$, and let $\mathfrak{p}$ be the associated prime ideal of the weak primary ideals $\mathfrak{q}_i$. Then $\mathfrak{p}$ also consists exactly of the elements of which some power is divisible by $\mathfrak{f}$. From $a \not\equiv 0 \pmod{\mathfrak{f}}$ and $b^x \not\equiv 0 \pmod{\mathfrak{f}}$ for every x it follows that $b \not\equiv 0 \pmod{\mathfrak{p}}$ and $a \not\equiv 0 \pmod{\mathfrak{q}_i}$ for at least one index i; hence $ab \not\equiv 0 \pmod{\mathfrak{q}_i}$, and therefore $ab \not\equiv 0 \pmod{\mathfrak{f}}$. If one replaces elements throughout by ideals, one can no longer infer $\mathfrak{b} \not\equiv 0 \pmod{\mathfrak{p}}$.

Now let $\mathfrak{m} = [\mathfrak{q}_1, \ldots, \mathfrak{q}_r]$ be a shortest representation by weak primary ideals with $\mathfrak{p}_i \neq \mathfrak{p}_k$ for $i \neq k$, and let $r \geq 2$. There is then at least one associated prime ideal, say $\mathfrak{p}_1$, which is not contained in any of the others; for every proper multiple-chain of the $\mathfrak{p}$'s must terminate after at most r steps. Hence there are elements a_i of $\mathfrak{p}_i$ such that $a_i^{\varrho_i} \equiv 0 \pmod{\mathfrak{q}_i}$, while $a_2 \not\equiv 0 \pmod{\mathfrak{p}_1}, \ldots, a_r \not\equiv 0 \pmod{\mathfrak{p}_1}$. If further $q_1 \not\equiv 0 \pmod{\mathfrak{m}}$ is

[21] In *Idealtheorie*, §4, axiom I, the divisor-chain condition, is assumed; therefore it is not made sharp there which properties of prime and primary ideals are independent of this axiom.

[22] This is shown by the following example communicated to me by R. Hölzer. Let $\mathfrak{R}$ be the polynomial domain in countably many indeterminates x_i with coefficients in a field, and let $\mathfrak{q} = (x_1^2, x_2^3, \ldots, x_\nu^{\nu+1}, \ldots, x_i x_k\ [i \neq k] \ldots)$. The zero divisors in the residue-class ring are exactly the polynomials divisible by $\mathfrak{p} = (x_1, x_2, \ldots, x_\nu, \ldots)$, and in each case a power of such a zero divisor is divisible by $\mathfrak{q}$; therefore $\mathfrak{q}$ is a weak primary ideal with associated prime ideal $\mathfrak{p}$. But $\mathfrak{q}$ is not a strong primary ideal: putting $\mathfrak{a} = (x_1, x_3, x_5, \ldots, x_{2\nu+1}, \ldots)$ and $\mathfrak{b} = (x_2, x_4, \ldots, x_{2\nu}, \ldots)$, one has $\mathfrak{ab} \equiv 0 \pmod{\mathfrak{q}}$, but no power of $\mathfrak{a}$ or $\mathfrak{b}$ is divisible by $\mathfrak{q}$.

[23] In order to infer the finite ideal basis from axiom I, a fixed well-ordering in $\mathfrak{R}$ must be used; compare §6.

an element of $\mathfrak{q}_1$, then $q_1 a_2^{\varrho_2} \cdots a_r^{\varrho_r} \equiv 0 \pmod{\mathfrak{m}}$, but no power of $a_2^{\varrho_2} \cdots a_r^{\varrho_r}$ is divisible by $\mathfrak{m}$, since otherwise divisibility by $\mathfrak{p}_1$ would follow. Thus $\mathfrak{m}$ is not a weak primary ideal. If elements are replaced throughout by ideals, that is q_1 by $\mathfrak{q}_1$ and a_i by $\mathfrak{a}_i[\not\equiv 0 \pmod{\mathfrak{p}_1}$, but $\mathfrak{a}_i^{\varrho_i} \equiv 0 \pmod{\mathfrak{q}_i}]$, the proof for strong primary ideals is obtained.[24]

4. *Module quotient and ideal quotient.* Let $\mathfrak{T}$ be an extension ring of $\mathfrak{R}$, and let $\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \ldots$ be $\mathfrak{R}$-modules in $\mathfrak{T}$. The quotient $\mathfrak{C} = \mathfrak{A} : \mathfrak{B}$ is defined as a module satisfying the conditions $\mathfrak{C}\mathfrak{B} \equiv 0 \pmod{\mathfrak{A}}$ and, if $\mathfrak{D}\mathfrak{B} \equiv 0 \pmod{\mathfrak{A}}$, then $\mathfrak{D} \equiv 0 \pmod{\mathfrak{C}}$; thus $\mathfrak{C}$ is the smallest module for which $\mathfrak{C}\mathfrak{B} \equiv 0 \pmod{\mathfrak{A}}$ holds. This determines the quotient uniquely as the greatest common divisor of all modules $\mathfrak{D}$ for which $\mathfrak{D}\mathfrak{B} \equiv 0 \pmod{\mathfrak{A}}$; such $\mathfrak{D}$ exist, since the zero module is certainly one of them.

From the definition it follows: if $\mathfrak{B}$ is a multiple of $\overline{\mathfrak{B}}$, then $\mathfrak{A} : \overline{\mathfrak{B}}$ is a multiple of $\mathfrak{A} : \mathfrak{B}$. If $\mathfrak{A}$ is a multiple of $\overline{\mathfrak{A}}$, then $\mathfrak{A} : \mathfrak{B}$ is a multiple of $\overline{\mathfrak{A}} : \mathfrak{B}$. Also

$$\mathfrak{A} : \mathfrak{B}\mathfrak{C} = (\mathfrak{A} : \mathfrak{B}) : \mathfrak{C} = (\mathfrak{A} : \mathfrak{C}) : \mathfrak{B}.$$

If the extension ring $\mathfrak{T}$ coincides with $\mathfrak{R}$, the module quotient becomes the ideal quotient $\mathfrak{a} : \mathfrak{b}$; hence the preceding rules also hold here. Since $\mathfrak{a}\mathfrak{b} \equiv 0 \pmod{\mathfrak{a}}$, one also has $\mathfrak{a} \equiv 0 \pmod{\mathfrak{a} : \mathfrak{b}}$.

5. *If $\mathfrak{a} : \mathfrak{b} = \mathfrak{a}$, then $\mathfrak{b}$ is called prime to $\mathfrak{a}$.* Thus an ideal $\mathfrak{b}$ is prime to $\mathfrak{a}$ if and only if $\mathfrak{d}\mathfrak{b} \equiv 0 \pmod{\mathfrak{a}}$ always implies $\mathfrak{d} \equiv 0 \pmod{\mathfrak{a}}$. From the calculation rules under 4 it follows: if $\mathfrak{b}$ and $\mathfrak{c}$ are prime to $\mathfrak{a}$, then so are their product and their least common multiple. If $\mathfrak{b}$ is prime to $\mathfrak{a}$ and $\mathfrak{a}$ is prime to $\mathfrak{b}$, then $\mathfrak{a}$ and $\mathfrak{b}$ are called mutually prime. From §4, 4β, it follows that coprime ideals are also mutually prime.

§6. Ideal theory under the assumption of the divisor-chain condition

Let the basis be a commutative ring $\mathfrak{R}$ for which axiom I, the divisor-chain condition, is satisfied. In what follows assume also a fixed well-ordering of all elements of $\mathfrak{R}$. This also determines a well-ordering of all ideals of $\mathfrak{R}$. Indeed, the two assumptions imply at once that every ideal of $\mathfrak{R}$ possesses an ideal basis consisting of finitely many elements. Passing from the well-ordering of the elements of $\mathfrak{R}$ to a quasi-lexicographic well-ordering of all finite subsets of $\mathfrak{R}$, [26] and assigning to each ideal, as distinguished basis, the first among its possible bases in this well-ordering of finite subsets, one obtains a well-ordering of the ideals together with that of the finite subsets.

Theorem I. *Under the assumption of the divisor-chain condition, every ideal of $\mathfrak{R}$ admits a representation as the least common multiple of finitely many irreducible ideals, i.e. of ideals which cannot be represented as the least common multiple of two proper divisors.*

For the proof it is enough to show: if Theorem I is false for an ideal $\mathfrak{m}$, then $\mathfrak{m}$ possesses a proper divisor for which Theorem I is likewise false; from this one can construct, contrary to the divisor-chain condition, a divisor chain not terminating after finitely many steps. Indeed $\mathfrak{m}$ must be reducible, since otherwise $\mathfrak{m} = [\mathfrak{m}]$ would already give the desired representation. If $\mathfrak{m} = [\mathfrak{a}, \mathfrak{b}]$, Theorem I cannot hold for both $\mathfrak{a}$ and $\mathfrak{b}$ at the same time; otherwise the corresponding representation would follow for $\mathfrak{m}$. Hence there are proper divisors of $\mathfrak{m}$ for which Theorem I is false; let $\mathfrak{a}_1$ be the first of these in the well-ordering of ideals. Constructing in the same way, from $\mathfrak{a}_1$, a proper divisor $\mathfrak{a}_2$, and in general from $\mathfrak{a}_i$ a proper divisor $\mathfrak{a}_{i+1}$, one obtains a well-defined divisor chain which does not terminate, contradicting the assumption. [27]

Because the divisor-chain condition is assumed, §5, 2 permits us simply to speak of primary ideals. The connection between primary and irreducible ideals is the following.

Theorem II. *Under the assumption of the divisor-chain condition, every irreducible ideal is primary; equivalently, every non-primary ideal is reducible.*

[24] I owe this modification of my original proof, which makes the divisor-chain condition unnecessary, to B. L. van der Waerden.

[26] This is to be understood as follows. If the elements $a_1, \ldots, a_n$ of a finite subset $\mathfrak{W}$ are denoted so that the ordering of the indices agrees with the prescribed well-ordering in $\mathfrak{R}$, then every subset with n elements precedes one with $m > n$ elements. If two subsets have the same number of elements, the first precedes the second when the first element at which their ordered lists differ precedes the corresponding element of the other list. Thus every system of finite subsets has a first element. Given a well-ordering of $\mathfrak{R}$, no further choice postulate is needed; in particular, if $\mathfrak{R}$ is countable, as in the case of an algebraic number field, no choice postulate is involved. The role of the choice postulate in ideal theory was mentioned, without further detail, in *Idealtheorie*, note 5.

[27] Theorem I plainly holds in the same way for module domains, if the divisor-chain condition is assumed for the system of all modules. The theorem has a purely set-theoretic character; it is the only place in the proof where the well-ordering of ideals is used. Abstractly: in a set M let a family Σ of subsets be distinguished and well-ordered. Assume the chain condition for Σ-sets: every chain $X_1, X_2, \ldots$ in which X_i is a proper superset of X_{i-1} terminates. A Σ-set is reducible if it is the intersection of two Σ-sets which are both proper supersets, otherwise irreducible. Then every Σ-set is an intersection of finitely many irreducible Σ-sets.

Passing to a residue-class ring $\mathfrak{R}/\mathfrak{m}$ preserves the divisor-chain condition by homomorphism. The representation of $\mathfrak{m}$ as a least common multiple corresponds to the representation of the zero ideal; and by the first isomorphism theorem (§4, 3), primary divisors of $\mathfrak{m}$ correspond again to primary ideals in $\mathfrak{R}/\mathfrak{m}$ and conversely. Thus it is enough to consider the decomposition of the zero ideal in the residue-class ring. Since that ring is again a ring of the same generality, one may from the outset suppose the ideal to be decomposed to be the zero ideal of $\mathfrak{R}$.

Suppose then that the zero ideal of $\mathfrak{R}$ is not primary. There is at least one pair of ideals $\mathfrak{a}, \mathfrak{b}$ – and $\mathfrak{b}$ may be assumed principal – such that

$$\mathfrak{a} \neq (0), \qquad \mathfrak{b}^x \neq (0) \quad \text{for every } x, \qquad \text{but} \qquad \mathfrak{a}\mathfrak{b} = (0).$$

Form the chain of ideal quotients

$$\mathfrak{a}, \quad \mathfrak{a} : \mathfrak{b}, \quad \ldots, \quad \mathfrak{a} : \mathfrak{b}^\nu, \quad \ldots .$$

This chain terminates; let, for instance,

$$\mathfrak{t} = \mathfrak{a} : \mathfrak{b}^{m-1} = \mathfrak{a} : \mathfrak{b}^m = \cdots .$$

Thus $\mathfrak{t} = \mathfrak{t} : \mathfrak{b}$, or $\mathfrak{b}$ is prime to $\mathfrak{t}$. Further, $\mathfrak{t}$ is a divisor of $\mathfrak{a}$ different from the zero ideal, and $\mathfrak{b}^x$ is different from zero for every exponent x. It therefore suffices, for the proof of Theorem II, to establish

$$(0) = [\mathfrak{t}, \mathfrak{b}^{m+1}].$$

By definition,

$$\mathfrak{b}^{m-1}\mathfrak{t} \equiv 0 \pmod{\mathfrak{a}}, \qquad \text{hence} \qquad \mathfrak{b}^m\mathfrak{t} = (0),$$

so it remains only to show

$$\mathfrak{c} = [\mathfrak{t}, \mathfrak{b}^{m+1}] \equiv 0 \pmod{\mathfrak{b}^m\mathfrak{t}}.$$

This follows from the assumptions that $\mathfrak{b}$ is principal and prime to $\mathfrak{t}$. Every element c of $\mathfrak{c}$, because of its divisibility by $\mathfrak{b}^{m+1}$, has a representation – where n denotes an integer symbol –

$$c = kb^{m+1} + nb^{m+1} = rb^m,$$

where r is again an element of $\mathfrak{R}$. From $c \equiv 0 \pmod{\mathfrak{t}}$ follows $rb^m \equiv 0 \pmod{\mathfrak{t}}$, and hence $r \equiv 0 \pmod{\mathfrak{t}}$, since $\mathfrak{t} : \mathfrak{b} = \mathfrak{t}$. Therefore $c \equiv 0 \pmod{\mathfrak{t}\mathfrak{b}^m}$, and Theorem II is proved.

Theorem III. *Under the assumption of the divisor-chain condition, every ideal admits a shortest representation as the least common multiple of finitely many greatest primary components belonging to distinct prime ideals.*

To obtain such a representation, replace, whenever possible, the representation by irreducible ideals furnished by Theorem I – hence, by Theorem II, by primary ideals – by a shortest representation. If the primary ideals belonging to the same prime ideal are collected together, §5, 3 gives the asserted representation. The primary components may be called greatest because the least common multiple of any of them is no longer primary. [28]

§7. Ideal theory under the assumption of the double-chain condition

The simplifications, compared with the theory developed so far, rest on the auxiliary results given under 1 and 2.

1. *If, in a commutative ring without zero divisors, the multiple-chain condition is satisfied, then the ring is already a field.* It is to be shown that, for $a \neq 0$, the equation $ax = b$ always has a solution in the ring. That it can have at most one solution follows from the absence of zero divisors.

Let $\mathfrak{a}$ be the principal ideal generated by a. By assumption the sequence of powers terminates; let $\mathfrak{a}^m$ be equal to all following powers. If $\mathfrak{b}$ denotes the principal ideal generated by b, then

$$\mathfrak{a}^m\mathfrak{b} = \mathfrak{a}^{m+1}\mathfrak{b}, \qquad \mathfrak{a}^{m+1}\mathfrak{b} = (a^{m+1}b).$$

[28] The corresponding prime ideals and the isolated components are uniquely determined. Compare *Idealtheorie*, or W. Krull, "Ein neuer Beweis für die Hauptsätze der allgemeinen Idealtheorie," Math. Ann. 90 (1923), pp. 55–64. In §7 a direct proof of uniqueness is given for the simple special case occurring there. The primary ideals there recognized as unique are isolated components.

In particular the element $a^m b$ has a representation – again with n an integer symbol –

$$a^m b = k a^{m+1} b + n a^{m+1} b = a^{m+1} c,$$

where c is an element of $\mathfrak{R}$. Since no zero divisors exist, this gives $b = ac$, proving that the ring is a field.

2. From 1 follows at once the auxiliary result: *If the multiple-chain condition is satisfied in a commutative ring, then a prime ideal has no proper divisor different from* $\mathfrak{o}$.

Likewise one obtains the supplement: *If only axiom II is satisfied, that is, the multiple-chain condition modulo every ideal different from the zero ideal, then every prime ideal different from the zero ideal has no proper divisor different from* $\mathfrak{o}$.

For the residue-class ring modulo a prime ideal $\mathfrak{p}$ is, by definition, a ring without zero divisors; since the multiple-chain condition is also preserved there by homomorphism, it is a field by 1. From the one-to-one correspondence between the divisors of $\mathfrak{p}$ and the ideals in the residue-class ring, $\mathfrak{p}$ has no proper divisor other than $\mathfrak{o}$. [29]

Theorem IV. *If the double-chain condition is satisfied in a commutative ring, then in every shortest representation of an ideal the primary components not belonging to the unit ideal* $\mathfrak{o}$ *are uniquely determined.*

Supplement. Under the assumptions of axioms I and II, Theorem IV holds for every ideal different from the zero ideal.

Let

$$\mathfrak{m} = [\mathfrak{q}, \mathfrak{q}_1, \ldots, \mathfrak{q}_r] = [\overline{\mathfrak{q}}, \overline{\mathfrak{q}}_1, \ldots, \overline{\mathfrak{q}}_s]$$

be shortest representations of $\mathfrak{m}$ by greatest primary components, with associated prime ideals

$$\mathfrak{o}, \mathfrak{p}_1, \ldots, \mathfrak{p}_r, \qquad \mathfrak{o}, \overline{\mathfrak{p}}_1, \ldots, \overline{\mathfrak{p}}_s.$$

The component $\mathfrak{q}$, respectively $\overline{\mathfrak{q}}$, is to be omitted when no primary ideal belonging to $\mathfrak{o}$ occurs. Since

$$\mathfrak{o}\,\mathfrak{p}_1 \cdots \mathfrak{p}_r \equiv 0 \pmod{\mathfrak{m}},$$

each $\mathfrak{p}_i$ must be contained in some $\overline{\mathfrak{p}}_j$, hence by the auxiliary result is identical with it. Conversely each $\overline{\mathfrak{p}}_j$ must coincide with some $\mathfrak{p}_i$; the prime ideals different from $\mathfrak{o}$ therefore agree. Further,

$$\mathfrak{q}\,\mathfrak{q}_1 \cdots \mathfrak{q}_r \equiv 0 \pmod{\mathfrak{q}_i}$$

implies $\overline{\mathfrak{q}}_i \equiv 0 \pmod{\mathfrak{q}_i}$, because the remaining components are not divisible by $\mathfrak{p}_i$ and hence are prime to $\mathfrak{q}_i$. The converse congruence follows in the same way, so the non-$\mathfrak{o}$ primary components are unique. If $\mathfrak{q}$ actually occurs, then $\overline{\mathfrak{q}}$ must actually occur as well, because the representations are shortest; this also gives the uniqueness of the associated prime ideals. The proof also shows that every representation without a primary component belonging to $\mathfrak{o}$ is already shortest.

3. If to the double-chain condition one adds the existence of an identity element, Theorem IV sharpens to the following.

Theorem V. *In a commutative ring with identity in which the double-chain condition is satisfied, every ideal can be represented uniquely as a product of finitely many pairwise coprime primary ideals.*

Supplement. Under axioms I, II, III, Theorem V holds for every ideal different from the zero ideal.

Because an identity element exists, $\mathfrak{o}^2 = \mathfrak{o}$; hence every primary ideal belonging to $\mathfrak{o}$ is equal to $\mathfrak{o}$ and cannot occur in a shortest representation of an ideal different from $\mathfrak{o}$. Thus, by Theorem IV, the primary components are uniquely determined, and at the same time every representation becomes shortest after omitting $\mathfrak{o}$. By the auxiliary result under 2, two distinct prime ideals are always coprime; by the calculation rules of §4, 4, the same holds for their primary components. The least common multiple therefore becomes their product.

From the same assumptions follows the auxiliary result: *In a commutative ring with identity and the double-chain condition, the prime ideals prime to an ideal* $\mathfrak{m}$ *are exactly those, apart from the unit ideal, which are not contained in* $\mathfrak{m}$. *The notions prime and coprime coincide.*

Supplement. Under axioms I, II, III, the ideal $\mathfrak{m}$ must be assumed different from the zero ideal.

[29] No identity element need exist in $\mathfrak{R}$, nor therefore in $\mathfrak{o}$, the ideal consisting of all elements of $\mathfrak{R}$, although an identity class exists in the residue-class ring by 1. The simplest example of this kind, in the weaker case of the supplement, is furnished by the system of all even integers.

If $\mathfrak{p}$ is not contained in $\mathfrak{m}$, then it is different from all prime ideals belonging to $\mathfrak{m}$, and by the auxiliary result under 2 it is not divisible by any of them. Hence it is prime to each primary component and therefore to $\mathfrak{m}$; at the same time it is coprime to each component and hence to $\mathfrak{m}$.

If, however, $\mathfrak{p} \neq \mathfrak{o}$ is contained in $\mathfrak{m}$, then it must coincide with one of the associated prime ideals. Let it belong to the component $\mathfrak{q} = \mathfrak{q}_1$. Then $\mathfrak{q} : \mathfrak{p}$ is a proper divisor of $\mathfrak{q}$, since

$$\mathfrak{p}^{\varrho-1} \equiv 0 \pmod{\mathfrak{q} : \mathfrak{p}}, \qquad \mathfrak{p}^{\varrho-1} \not\equiv 0 \pmod{\mathfrak{q}}.$$

At the same time $\mathfrak{q} : \mathfrak{p}$, as a divisor of $\mathfrak{p}^{\varrho-1}$, is divisible only by the prime ideal $\mathfrak{p} \neq \mathfrak{o}$ and is therefore primary. By the uniqueness of the decomposition, the product

$$(\mathfrak{q} : \mathfrak{p})\mathfrak{q}_2 \cdots \mathfrak{q}_r,$$

which is divisible by $\mathfrak{m} : \mathfrak{p}$, is a proper divisor of $\mathfrak{m}$; hence $\mathfrak{p}$ is not prime to $\mathfrak{m}$.

Thus an ideal $\mathfrak{t}$ is prime to $\mathfrak{m}$ if and only if none of the prime ideals belonging to $\mathfrak{t}$ is contained in $\mathfrak{m}$; but in that case $\mathfrak{t}$ is also coprime to $\mathfrak{m}$. Conversely, coprime ideals are always mutually prime (§5, 5), so the two concepts coincide under these assumptions.

§8. Ideal theory under integral closedness in the quotient field

The ideal theory developed so far is now to be sharpened to the usual one by adjoining the last two axioms.

1. *Auxiliary result.* Let $\mathfrak{R}$ be a commutative ring without zero divisors and with an identity element. If the divisor-chain condition is assumed in $\mathfrak{R}$ (axioms I, III, IV), then from $\mathfrak{a} = \mathfrak{a}\mathfrak{b}$ and $\mathfrak{a} \neq (0)$ it follows that $\mathfrak{b} = \mathfrak{o}$. Hence

$$\mathfrak{c} \neq \mathfrak{c}\mathfrak{t}$$

for all ideals different from the zero and unit ideals. [30] The proof is the same as for the auxiliary result §1, 3, since $\mathfrak{a}\mathfrak{b}$ may be regarded as a finite module over $\mathfrak{b}$. Let $\alpha_1, \ldots, \alpha_n$ be an ideal basis of $\mathfrak{a}$ existing by I. From $\mathfrak{a} = \mathfrak{a}\mathfrak{b}$ one obtains the system

$$\alpha_i = b_{i1}\alpha_1 + \cdots + b_{in}\alpha_n, \qquad b_{ij} \equiv 0 \pmod{\mathfrak{b}}, \quad i = 1, \ldots, n.$$

Since $\mathfrak{R}$ has no zero divisors, the determinant $|b_{ij} - e_{ij}|$ vanishes; with $e_{ij} = 0$ for $i \neq j$ and $e_{ii} = e$. From

$$e - b_{11} - \cdots \equiv 0 \pmod{\mathfrak{b}}$$

it follows that $e \equiv 0 \pmod{\mathfrak{b}}$ and therefore $\mathfrak{b} = \mathfrak{o}$.

2. It remains to show only that, after adjoining integral closedness in the quotient field (axiom V) to axioms I through IV, every primary ideal different from the zero ideal is a power of its associated prime ideal. The uniqueness of the resulting representation

$$\mathfrak{m} = \mathfrak{p}_1^{\varrho_1} \cdots \mathfrak{p}_r^{\varrho_r}$$

for every ideal different from the zero ideal has already been proved by Theorem V and the auxiliary result under 1. At the same time it has been shown that these prime ideals have no proper divisor distinct from the unit ideal. The proof is first given for exponent two, using Dedekind's corollary II (§1, 4), and then in general by complete induction.

Auxiliary result. Under axioms I through V there are no primary ideals of exponent two other than $\mathfrak{q} = \mathfrak{p}^2$.

It is to be shown that from

$$\mathfrak{p}^2 \equiv 0 \pmod{\mathfrak{q}}, \qquad \mathfrak{q} \equiv 0 \pmod{\mathfrak{p}}, \qquad \mathfrak{q} \not\equiv 0 \pmod{\mathfrak{p}^2}$$

there necessarily follows $\mathfrak{q} = \mathfrak{p}^2$. Choose

$$\mathfrak{c} \equiv 0 \pmod{\mathfrak{q}}, \qquad \mathfrak{c} \not\equiv 0 \pmod{\mathfrak{p}}, \qquad \mathfrak{c} \equiv 0 \pmod{\mathfrak{p}^2}.$$

From the auxiliary result §7, 3, it follows that $\mathfrak{c} : \mathfrak{p}$ is a proper divisor of $\mathfrak{c}$; hence there is an element b such that

$$b\mathfrak{p} \equiv 0 \pmod{\mathfrak{c}} \equiv 0 \pmod{\mathfrak{q}}, \qquad b \not\equiv 0 \pmod{\mathfrak{c}}.$$

[30] On the other hand, under the same assumptions one cannot infer $\mathfrak{b} = \mathfrak{c}$ from $\mathfrak{a}\mathfrak{b} = \mathfrak{a}\mathfrak{c}$. For example, let $\mathfrak{R}$ be the polynomial domain in x, y over a field, with $\mathfrak{a} = (xy)$, $\mathfrak{b} = (x^3, xy, y^2)$, $\mathfrak{c} = (x^2, y^2)$. Then $\mathfrak{a}\mathfrak{b} = \mathfrak{a}\mathfrak{c} = \mathfrak{a}^2$, but $\mathfrak{b} \neq \mathfrak{c}$.

Thus $y = b/c$ belongs to the quotient field but not to $\mathfrak{R}$; it is not integral. We must prove $b \not\equiv 0 \pmod{\mathfrak{p}}$; then, since $\mathfrak{q}$ is primary and belongs to $\mathfrak{p}$, it follows that $\mathfrak{p} \equiv 0 \pmod{\mathfrak{q}}$.

By Dedekind's corollary II there exist elements m, n of $\mathfrak{R}$ such that

$$y = b/c = m/n, \qquad m^2/n \text{ is not integral;}$$

that is,

$$bn = mc, \qquad m \not\equiv 0 \pmod{\mathfrak{n}}, \qquad m^2 \not\equiv 0 \pmod{\mathfrak{n}}.$$

Multiplying $b\mathfrak{p}$ by m gives

$$mb\mathfrak{p} \equiv 0 \pmod{\mathfrak{n}c}, \qquad \text{hence} \qquad m\mathfrak{p} \equiv 0 \pmod{\mathfrak{n}}.$$

Therefore $m \not\equiv 0 \pmod{\mathfrak{p}}$, because $m^2 \not\equiv 0 \pmod{\mathfrak{n}}$ and $\mathfrak{n} \not\equiv 0 \pmod{\mathfrak{p}}$; the latter again follows from §7, 3, since $\mathfrak{n} : \mathfrak{p}$ is a proper divisor of $\mathfrak{n}$. The four relations

$$m \not\equiv 0 \pmod{\mathfrak{p}}, \qquad n \not\equiv 0 \pmod{\mathfrak{p}}, \qquad c \equiv 0 \pmod{\mathfrak{p}}, \qquad c \not\equiv 0 \pmod{\mathfrak{p}^2}, \qquad bn = mc$$

now imply $b \not\equiv 0 \pmod{\mathfrak{p}}$. For, since $\mathfrak{p}^2$ is primary, one first gets $mc \not\equiv 0 \pmod{\mathfrak{p}^2}$, and then $b \not\equiv 0 \pmod{\mathfrak{p}}$ from $bn = mc$. This proves the auxiliary result.

3. *Auxiliary result. Under axioms I through V there are no primary ideals of exponent ϱ other than $\mathfrak{q} = \mathfrak{p}^\varrho$.*

This follows from the auxiliary result under 2 without further use of the axioms. [31] First one proves:

$$\text{if } \mathfrak{c} \equiv 0 \pmod{\mathfrak{p}}, \ \mathfrak{c} \not\equiv 0 \pmod{\mathfrak{p}^2}, \quad \text{then} \quad \mathfrak{p}^\varrho = (\mathfrak{c}^\varrho, \mathfrak{p}^{\varrho+1})$$

for every ϱ. By the result under 2, $\mathfrak{p} = (\mathfrak{c}, \mathfrak{p}^2)$. Assuming $\mathfrak{p}^{\varrho-1} = (\mathfrak{c}^{\varrho-1}, \mathfrak{p}^\varrho)$, multiplication by $\mathfrak{p}$ and the distributive law give

$$\mathfrak{p}^\varrho = (\mathfrak{c}^\varrho, \mathfrak{p}^{\varrho+1}).$$

The desired result may be put in the following second form: if

$$\mathfrak{q} \equiv 0 \pmod{\mathfrak{p}^\varrho}, \qquad \mathfrak{q} \not\equiv 0 \pmod{\mathfrak{p}^{\varrho+1}}, \qquad \mathfrak{p}^{\varrho+1} \equiv 0 \pmod{\mathfrak{q}}, \qquad \varrho \geq 1,$$

then already $\mathfrak{p}^\varrho \equiv 0 \pmod{\mathfrak{q}}$. Indeed, for every primary ideal there is such an exponent $\varrho > 1$; the condition $\mathfrak{q} \not\equiv 0 \pmod{\mathfrak{p}^{\varrho+1}}$ follows from $\mathfrak{p}^\varrho \equiv 0 \pmod{\mathfrak{q}}$ and $\mathfrak{p}^\varrho \neq \mathfrak{p}^{\varrho+1}$ by the result under 1. Repeated application of the second form gives $\mathfrak{p}^\varrho \equiv 0 \pmod{\mathfrak{q}}$ and hence $\mathfrak{q} = \mathfrak{p}^\varrho$.

From the assumption of the second form and from the representation of $\mathfrak{p}^\varrho$, each element q of $\mathfrak{q}$ can be written

$$q = bc^\varrho + p^{\varrho+1}, \qquad b \equiv 0 \pmod{\mathfrak{p}}, \qquad \text{or} \qquad bc^\varrho \equiv 0 \pmod{\mathfrak{q}, \mathfrak{p}^{\varrho+1}}.$$

Since $(\mathfrak{q}, \mathfrak{p}^{\varrho+1})$ is primary, it follows that

$$c^\varrho \equiv 0 \pmod{\mathfrak{q}, \mathfrak{p}^{\varrho+1}}, \qquad \text{or} \qquad c^{\varrho+1} \equiv 0 \pmod{\mathfrak{q}, \mathfrak{p}^{\varrho+2}},$$

and hence $\mathfrak{p}^\varrho \equiv 0 \pmod{\mathfrak{q}}$.

4. Summarizing, one obtains:

Theorem VI. *If axioms I through V hold in a commutative ring, then every ideal different from the zero and unit ideals can be represented uniquely as a product of powers of finitely many prime ideals different from the zero and unit ideals. These prime ideals have no proper divisor different from the unit ideal.*

§9. The axioms as consequences of the assumed decomposition

Conversely, in order to infer the axioms from the existence of the usual ideal decomposition – which by Theorem VI follows from axioms I through V – the decomposition must be assumed in the following form.

Assumption. In the commutative ring $\mathfrak{R}$, every ideal not generated by the zero or the unit element is representable uniquely as a product of powers of prime ideals. These prime ideals are simple ideals not generated by the unit element, i.e. they have no proper divisor different from $\mathfrak{o}$; conversely all simple ideals are prime ideals. The uniqueness is to hold in the sharp form that

$$\mathfrak{a}^\varrho \neq \mathfrak{a}^{\varrho+1}$$

[31] The proof of auxiliary result 3 under the assumption of auxiliary result 2 is found in Masazo Sono, *On Congruences II*, §§9 and 10. Compare note 17.

for every ideal not generated by the zero or by the unit element.

The existence of an identity element is not assumed here. If no identity element exists, the condition “not generated by the unit element” is no restriction.

1. *Proof of axiom III, the existence of the identity element.* Suppose axiom III is not satisfied. It is to be shown that $\mathfrak{o}^2$ becomes a simple ideal without being prime, contrary to the assumption. By the sharp uniqueness assumption, $\mathfrak{o} \neq \mathfrak{o}^2$; hence $\mathfrak{o} \equiv 0 \pmod{\mathfrak{o}^2}$, but $\mathfrak{o}^2 \not\equiv 0 \pmod{\mathfrak{o}^2}$, and so $\mathfrak{o}^2$ is not prime. It is, however, simple: it cannot be divisible by any prime ideal different from $\mathfrak{o}$, and by the assumed product representation it has no divisors except $\mathfrak{o}$ and $\mathfrak{o}^2$.

2. *Proof of axiom IV, absence of zero divisors.* If a is a zero divisor, say $ab = 0$ with $a \neq 0$ and $b \neq 0$, then multiplication of the product representations of the principal ideals generated by a and b gives

$$(0) = \mathfrak{p}_1^{\varrho_1} \cdots \mathfrak{p}_r^{\varrho_r},$$

where the $\mathfrak{p}_i$ are different from the zero and unit ideals, the latter because the identity element has already been proved to exist. Take the representation to be shortest, in the sense that deleting any factor yields a proper divisor of the zero ideal; this need not be automatic, since no uniqueness assumption has been made about the zero ideal. If only one prime ideal occurs, then $(0) = \mathfrak{p}^{\varrho} = \mathfrak{p}^{\varrho+1}$, contrary to the sharp uniqueness demand. If more than one prime ideal occurs, then

$$\begin{aligned} \mathfrak{p}_1^{\varrho_1} &= (\mathfrak{p}_1^{\varrho_1}, \mathfrak{p}_2^{\varrho_2} \cdots \mathfrak{p}_r^{\varrho_r})\mathfrak{p}_1^{\varrho_1} \\ &= \mathfrak{p}_1^{\varrho_1}\mathfrak{p}_2^{\varrho_2} \cdots \mathfrak{p}_r^{\varrho_r}, \end{aligned}$$

because, by the existence of the identity element, $\mathfrak{p}_1$ is coprime to all the other prime ideals. This again contradicts the sharp uniqueness condition.

3. *Proof of axioms I and II, the chain conditions.* The assumptions give, for every ideal different from the zero ideal: divisibility is reflected in the product representation. Thus from $\mathfrak{m} \equiv 0 \pmod{\mathfrak{a}}$ and $\mathfrak{a} \neq \mathfrak{o}$, with

$$\mathfrak{m} = \mathfrak{p}_1^{\varrho_1} \cdots \mathfrak{p}_r^{\varrho_r}, \qquad \mathfrak{a} = \bar{\mathfrak{p}}_1^{\bar{\varrho}_1} \cdots \bar{\mathfrak{p}}_s^{\bar{\varrho}_s},$$

it follows that every $\bar{\mathfrak{p}}_j$ is identical with some $\mathfrak{p}_i$ and that $0 \leq \bar{\varrho}_i \leq \varrho_i$. Hence there are only finitely many divisors of $\mathfrak{m}$, corresponding to the finitely many exponent combinations. Therefore the double-chain condition holds in the residue-class ring modulo $\mathfrak{m}$. Axioms I and II are thus proved: the divisor-chain condition also holds in $\mathfrak{R}$ itself, since every proper divisor of the zero ideal is different from the zero ideal.

The proof of axiom V, integral closedness in the quotient field, uses the standard consequences of ideal decomposition, which must therefore first be derived.

4. *The residue-class ring modulo every ideal different from the zero ideal is a principal-ideal ring.* The ideal may be assumed different from the unit ideal, since for the residue-class ring consisting only of the zero element the assertion is immediate.

If the ideal is first primary, $\mathfrak{q} = \mathfrak{p}^{\varrho}$, and if $\mathfrak{c} \equiv 0 \pmod{\mathfrak{p}}$ but $\mathfrak{c} \not\equiv 0 \pmod{\mathfrak{p}^2}$, then by 3 one has $\mathfrak{c} = \mathfrak{p}\mathfrak{a}$ with $(\mathfrak{a}, \mathfrak{p}) = \mathfrak{o}$. Consequently

$$\mathfrak{p}^{\sigma} = (\mathfrak{c}^{\sigma}, \mathfrak{p}^{\sigma+1}) \qquad (\sigma < \varrho),$$

so every ideal in the residue-class ring modulo a primary ideal is principal. In general, if

$$\mathfrak{m} = \mathfrak{q}_1 \cdots \mathfrak{q}_r$$

is the representation and

$$\mathfrak{R}/\mathfrak{m} = \mathfrak{a}_1 + \cdots + \mathfrak{a}_r$$

the corresponding representation of the residue-class ring as a direct sum (§4, 5), then $\mathfrak{a}_i$ is isomorphic to $\mathfrak{R}/\mathfrak{q}_i$; every ideal of $\mathfrak{a}_i$ is therefore principal. If $\mathfrak{c}$ is any ideal of the residue-class ring, then $\mathfrak{a}_i\mathfrak{c}$ is a principal ideal $\mathfrak{a}_i c_i$, and

$$\mathfrak{c} = \mathfrak{a}_1 c_1 + \cdots + \mathfrak{a}_r c_r = \mathfrak{o}c, \qquad c = c_1 + \cdots + c_r.$$

Indeed $\mathfrak{a}_i\mathfrak{o}c_i$ and $\mathfrak{a}_i\mathfrak{c}$ agree in $\mathfrak{a}_i$, since $\mathfrak{a}_i\mathfrak{a}_k = (0)$ for $i \neq k$ and $c_i \equiv 0 \pmod{\mathfrak{a}_i}$.

The theorem on principal-ideal residue rings also has the following form. If $\mathfrak{c}$ is any ideal, it can be transformed into a principal ideal by multiplication with an ideal prime to a prescribed ideal $\mathfrak{b}$. For, putting $\mathfrak{m} = \mathfrak{b}\mathfrak{c}$, the ideal $\mathfrak{c}$ becomes principal modulo $\mathfrak{m}$; that is,

$$\mathfrak{c} = (\mathfrak{o}c, \mathfrak{m}) = (c\mathfrak{a}, \mathfrak{c}\mathfrak{b}) = \mathfrak{c}(\mathfrak{a}, \mathfrak{b}),$$

so that $\mathfrak{o}c = \mathfrak{c}\mathfrak{a}$ and $(\mathfrak{a}, \mathfrak{b}) = \mathfrak{o}$.

5. *Theory of fractional ideals.* A fractional ideal means a finite $\mathfrak{R}$-module in the quotient field $\mathfrak{K}$.

The nonzero finite $\mathfrak{R}$-modules in $\mathfrak{K}$ form an abelian group under multiplication.

The product of two finite $\mathfrak{R}$-modules is again finite; multiplication is associative and commutative; since $\mathfrak{K} = \mathfrak{o}\mathfrak{K}$, the unit ideal is the identity element. It remains only to show that the equation

$$\mathfrak{U}X = \mathfrak{o}$$

always has a solution in the system of finite $\mathfrak{R}$-modules.

Preliminary remark: if the equation $\mathfrak{A}T = \mathfrak{B}$ has one and only one solution, then T is the module quotient $\mathfrak{B} : \mathfrak{A}$ (§5, 4). For

$$T \equiv 0 \pmod{\mathfrak{B} : \mathfrak{A}}, \qquad \mathfrak{B} = \mathfrak{A}T \equiv 0 \pmod{\mathfrak{A}(\mathfrak{B} : \mathfrak{A})} \equiv 0 \pmod{\mathfrak{B}}.$$

If first $\mathfrak{A}$ is principal, $\mathfrak{A} = (\alpha)$, then $X = (e/\alpha)$ is a solution of $\mathfrak{A}X = \mathfrak{o}$, and X is again a finite $\mathfrak{R}$-module. It follows that every finite $\mathfrak{R}$-module is the module quotient of two ideals of $\mathfrak{R}$, which justifies the term fractional ideal. Indeed, if $t_1, \ldots, t_n$ is a module basis of T, put $t_i = \tau_i/a$ and let $\mathfrak{t}$ be the ideal generated by the τ_i in $\mathfrak{R}$. Then $\mathfrak{o}a\,T = \mathfrak{t}$, whence by the preliminary remark $T = (\mathfrak{t} : \mathfrak{o}a)$, the quotient being taken in $\mathfrak{K}$.

The solution of $\mathfrak{A}X = \mathfrak{o}$ can now be given generally. Put $\mathfrak{A} = \mathfrak{a} : \mathfrak{o}c$, and choose $\mathfrak{b}$ so that $\mathfrak{a}\mathfrak{b}$ is the principal ideal $\mathfrak{o}a$. Then

$$\mathfrak{A}\,\mathfrak{o}c = \mathfrak{a}, \qquad \mathfrak{A}\,\mathfrak{b}\mathfrak{o}c = \mathfrak{o}a, \qquad \mathfrak{A}(\mathfrak{b}\mathfrak{o}c : \mathfrak{o}a) = \mathfrak{o}.$$

Here X, as the quotient of an ideal by a principal ideal, is again a finite $\mathfrak{R}$-module. The group property is proved. It also follows that the module quotient of any two ideals, $\mathfrak{a} : \mathfrak{b}$, is a finite $\mathfrak{R}$-module, as the solution of $\mathfrak{b}X = \mathfrak{a}$.

6. From the group property follow further consequences.

6 α. *Cancellation is possible: from $\mathfrak{T} = \mathfrak{A}\mathfrak{M} : \mathfrak{B}\mathfrak{M}$ follows $\mathfrak{T} = \mathfrak{A} : \mathfrak{B}$, and conversely.* For $\mathfrak{T}\mathfrak{B}\mathfrak{M} = \mathfrak{A}\mathfrak{M}$ gives $\mathfrak{T}\mathfrak{B} = \mathfrak{A}$, and conversely.

6 β. *Every fractional ideal admits a representation $\mathfrak{C} = \mathfrak{a} : \mathfrak{b}$, where $\mathfrak{a}$ and $\mathfrak{b}$ are coprime; this condition determines $\mathfrak{a}$ and $\mathfrak{b}$ uniquely.* Existence follows from 5 and 6 α. From $\mathfrak{C} = \mathfrak{a} : \mathfrak{b} = \bar{\mathfrak{a}} : \bar{\mathfrak{b}}$ follows $\mathfrak{a}\bar{\mathfrak{b}} = \mathfrak{b}\bar{\mathfrak{a}}$, and hence uniqueness when both pairs $\mathfrak{a}, \mathfrak{b}$ and $\bar{\mathfrak{a}}, \bar{\mathfrak{b}}$ are assumed coprime.

6 γ. *Every principal module generated by a nonintegral element, i.e. by an element of $\mathfrak{K}$ not in $\mathfrak{R}$, admits a representation as a quotient of principal ideals such that no power of the numerator is divisible by the denominator.* Let the reduced representation be $\mathfrak{o}n = \mathfrak{b} : \mathfrak{c}$, so $(\mathfrak{b}, \mathfrak{c}) = \mathfrak{o}$; choose $\mathfrak{a}$ according to 4 so that $\mathfrak{o}b = \mathfrak{b}\mathfrak{a}$ and $(\mathfrak{a}, \mathfrak{c}) = \mathfrak{o}$. Then

$$\mathfrak{o}n = \mathfrak{a}\mathfrak{b} : \mathfrak{a}\mathfrak{c} = \mathfrak{o}b : \mathfrak{a}\mathfrak{c},$$

and since $\mathfrak{a}\mathfrak{c} = \mathfrak{o}b : \mathfrak{o}n$, the ideal $\mathfrak{a}\mathfrak{c}$ is principal as well; write

$$\mathfrak{o}n = \mathfrak{o}b : \mathfrak{o}c.$$

No power of b is divisible by c, because $(\mathfrak{a}, \mathfrak{c}) = \mathfrak{o}$ and $(\mathfrak{b}, \mathfrak{c}) = \mathfrak{o}$.

7. *Proof of axiom V, integral closedness in the quotient field.* By 6 γ, every nonintegral element of $\mathfrak{K}$ has a quotient representation

$$\eta = a/c$$

such that, in $\mathfrak{R}$, no power of the numerator is divisible by the denominator. Such an element cannot satisfy an equation characterizing it as integral over $\mathfrak{R}$; for from

$$\eta^n + r_1\eta^{n-1} + \cdots + r_n = 0$$

one would obtain $a^n \equiv 0 \pmod{\mathfrak{o}c}$ in $\mathfrak{R}$.

8. *Consequence.* If axioms I through IV hold in a commutative ring $\mathfrak{R}$, and every primary ideal is irreducible, then axiom V, integral closedness in the quotient field, also holds. Because of axioms I through III, every prime-ideal power $\mathfrak{p}^\varrho$ is primary; in particular $\mathfrak{p}^2$ is primary and therefore irreducible by assumption. From this one must prove

$$\mathfrak{p} = (\mathfrak{o}c, \mathfrak{p}^2).$$

Then, by the auxiliary result §8, 3, every primary ideal is a prime-ideal power; together with the other assumptions, 7 gives integral closedness.

By the divisor-chain condition,

$$\mathfrak{p} = (\mathfrak{o}c_1, \dots, \mathfrak{o}c_n),$$

where only residue classes modulo $\mathfrak{p}$ are relevant as multipliers of the c_i, and the c_i may be assumed linearly independent over the residue-class field modulo $\mathfrak{p}$. If $n > 1$, this linear independence gives a representation

$$\mathfrak{p}^2 = [\mathfrak{q}_1, \mathfrak{q}_2], \qquad \mathfrak{q}_1 = (\mathfrak{o}c_1, \mathfrak{p}^2), \qquad \mathfrak{q}_2 = (\mathfrak{o}c_2, \dots, \mathfrak{o}c_n, \mathfrak{p}^2),$$

with both $\mathfrak{q}_1$ and $\mathfrak{q}_2$ proper divisors of $\mathfrak{p}^2$; hence $\mathfrak{p}^2$ would be reducible, contrary to the assumption. Thus $\mathfrak{p} = (\mathfrak{o}c, \mathfrak{p}^2)$, and the asserted consequence is proved.

Conversely, as a consequence of axioms I through V, every primary ideal is immediately irreducible: a prime-ideal power $\mathfrak{p}^\varrho$ cannot be the least common multiple of proper divisors of the form $\mathfrak{p}^\sigma$ and $\mathfrak{p}^\tau$.

9. If zero divisors are allowed in the ring, then axioms I through III and integral closedness in the quotient ring do not imply that every primary ideal is irreducible. [32] Let $\mathfrak{T}$ be a ring in which all axioms I through IV except integral closedness hold; such rings exist, for instance the finite orders distinct from the full system of integral quantities in a finite extension field of the quotient field of $\mathfrak{R}$, when axioms I through V hold in $\mathfrak{R}$ (§3, 2). By the consequence under 8 there is in $\mathfrak{T}$ a reducible primary ideal $\mathfrak{q}$. Let $\mathfrak{p}^\varrho$ be a proper multiple of $\mathfrak{q}$, and let $\mathfrak{R}$ be the residue-class ring $\mathfrak{T}/\mathfrak{p}^\varrho$, in which the ideal corresponding to $\mathfrak{q}$ is a nonzero reducible primary ideal. In $\mathfrak{R}$ axioms I through III hold, and integral closedness in the quotient ring also holds. For every element of $\mathfrak{T}$ not divisible by $\mathfrak{p}$ becomes coprime to $\mathfrak{p}^\varrho$; hence all regular elements of $\mathfrak{R}$ are units, and $\mathfrak{R}$ coincides with its quotient ring.

§10. The double-chain condition and composition series

We now show that, for arbitrary module domains (§2) assumed to be well-ordered, the validity of the double-chain condition – every divisor chain and every multiple chain of modules terminates after finitely many steps – is equivalent to the existence of a composition series. [33] Specializing the module domain to a commutative ring gives the corresponding facts for the system of all ideals of the ring; in 2, throughout, module isomorphism is then to be replaced by ring isomorphism.

A module domain $\mathfrak{G}$ is called *simple* if in $\mathfrak{G}$ there are no $\mathfrak{R}$-modules besides $\mathfrak{G}$ itself and the zero module $\mathfrak{E}$. A module $\mathfrak{A}$ is called *simple in* $\mathfrak{G}$ if $\mathfrak{G}/\mathfrak{A}$ is simple.

A multiple chain

$$\mathfrak{G}, \mathfrak{U}_1, \dots, \mathfrak{U}_r, \mathfrak{E}$$

is called a composition series of $\mathfrak{G}$ of length r if all modules in the chain are distinct and if each module is simple in the preceding one; equivalently, if the residue-class modules, the quotient groups, $\mathfrak{U}_{i-1}/\mathfrak{U}_i$ are simple module domains, as are $\mathfrak{G}/\mathfrak{U}_1$ and $\mathfrak{U}_r/\mathfrak{E}$. By forming the residue-class module $\mathfrak{G}/\mathfrak{F}$, the case in which a composition series exists from $\mathfrak{G}$ down to $\mathfrak{F} \neq \mathfrak{G}$ is reduced to the absolute case.

1. *If the double-chain condition is assumed in $\mathfrak{G}$, then a composition series exists in* $\mathfrak{G}$, provided $\mathfrak{G} \neq \mathfrak{E}$. From the assumed well-ordering of $\mathfrak{G}$ and the divisor-chain condition, one obtains as in §6, by quasi-lexicographic ordering, a well-ordering of the system of all modules. This well-ordering, together with the divisor-chain condition, gives the existence of at least one module simple in $\mathfrak{G}$. Namely, let $\mathfrak{G}_1$ be the first proper divisor of $\mathfrak{G}$ in the well-ordering; generally let $\mathfrak{G}_i$ be the first proper divisor of $\mathfrak{G}_{i-1}$. The well-determined chain

$$\mathfrak{G}, \mathfrak{G}_1, \dots, \mathfrak{G}_i, \dots$$

terminates after finitely many steps and therefore leads necessarily to a module $\mathfrak{A}$ simple in $\mathfrak{G}$. If $\mathfrak{U}_1 \neq \mathfrak{E}$, the same procedure gives a module $\mathfrak{U}_2$ simple in $\mathfrak{U}_1$; in general, if $\mathfrak{U}_{i-1} \neq \mathfrak{E}$, there is a module $\mathfrak{U}_i$ simple in $\mathfrak{U}_{i-1}$. This gives a well-determined multiple chain

$$\mathfrak{G}, \mathfrak{U}_1, \dots, \mathfrak{U}_i,$$

which terminates by the multiple-chain condition and is therefore a composition series of $\mathfrak{G}$.

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[32] Compare note 18.

[33] The modules of the domain are abelian groups with respect to addition, in fact generalized abelian groups, since the multiplier domain consists of the elements of $\mathfrak{R}$. Since they are abelian groups, the concept of composition series coincides here with that of principal series. The theorems of this paragraph remain valid if, under "modules," one understands the totality of normal divisors of an arbitrary noncommutative, even generalized, group. The notation, slightly different from the earlier one, is meant to recall group theory.

2. *If a composition series exists in $\mathfrak{G}$, then the double-chain condition holds in $\mathfrak{G}$.* The proof is by complete induction and does not require a well-ordering of $\mathfrak{G}$. In the course of the induction one obtains at the same time a proof of the Jordan–Hölder theorem under the sole assumption that a composition series exists. [34]

If $\mathfrak{G}$ is simple, so that $r = 0$ and the only composition series is $\mathfrak{G}, \mathfrak{E}$, then the following assertions hold.

α) Through every module simple in $\mathfrak{G}$ one can draw a composition series.

β) Jordan–Hölder theorem: every composition series has the same length and the system of quotient groups (residue-class modules) agrees up to order; corresponding quotient groups are isomorphic.

γ) Through every module different from $\mathfrak{G}$ one can draw a composition series.

δ) The multiple-chain condition holds.

ε) The divisor-chain condition holds.

Assume therefore that α)–ε) have been proved for every module domain possessing a composition series of length $r < r+1$. We prove them under the assumption that in $\mathfrak{G}$ there is a composition series

$$\mathfrak{G}, \mathfrak{A}, \mathfrak{A}_1, \ldots, \mathfrak{A}_r, \mathfrak{E}$$

of length $r+1$.

2α. If there is no module simple in $\mathfrak{G}$ other than $\mathfrak{A}$, there is nothing to prove. Let $\mathfrak{B}$ be simple in $\mathfrak{G}$ and $\mathfrak{B} \neq \mathfrak{A}$. Since $\mathfrak{A}$ and $\mathfrak{B}$ are both simple in $\mathfrak{G}$ and are distinct, necessarily $(\mathfrak{A}, \mathfrak{B}) = \mathfrak{G}$. Put $\mathfrak{M} = [\mathfrak{A}, \mathfrak{B}]$. By the second isomorphism theorem (§4, 2),

$$\mathfrak{G}/\mathfrak{B} \cong \mathfrak{A}/\mathfrak{M}, \qquad \mathfrak{G}/\mathfrak{A} \cong \mathfrak{B}/\mathfrak{M}.$$

Thus $\mathfrak{M}$ is simple in $\mathfrak{A}$ and in $\mathfrak{B}$. Since $\mathfrak{A}$ has a composition series of length $r < r+1$, the induction hypotheses α and δ give in $\mathfrak{A}$ a composition series through $\mathfrak{M}$, say

$$\mathfrak{A}, \mathfrak{M}, \mathfrak{M}_1, \ldots, \mathfrak{M}_{r-1}, \mathfrak{E}.$$

Consequently

$$\mathfrak{G}, \mathfrak{B}, \mathfrak{M}, \mathfrak{M}_1, \ldots, \mathfrak{M}_{r-1}, \mathfrak{E}$$

is a composition series of $\mathfrak{G}$ passing through $\mathfrak{B}$.

2β. To prove the Jordan–Hölder theorem, let

$$(1) \quad \mathfrak{G}, \mathfrak{A}, \mathfrak{A}_1, \ldots, \mathfrak{A}_r, \mathfrak{E}, \qquad \text{and} \qquad (2) \quad \mathfrak{G}, \mathfrak{B}, \mathfrak{B}_1, \ldots, \mathfrak{B}_s, \mathfrak{E}$$

be two composition series of $\mathfrak{G}$. If $\mathfrak{A} = \mathfrak{B}$, the result follows from the induction assumption for length $r < r+1$. Otherwise compare with the series constructed under 2α,

$$(3) \quad \mathfrak{G}, \mathfrak{B}, \mathfrak{M}, \mathfrak{M}_1, \ldots, \mathfrak{M}_{r-1}, \mathfrak{E}, \qquad (4) \quad \mathfrak{G}, \mathfrak{A}, \mathfrak{M}, \mathfrak{M}_1, \ldots, \mathfrak{M}_{r-1}, \mathfrak{E}.$$

By the induction hypothesis, the system of quotient groups of (1) and (3) agrees; likewise that of (2) and (4), since (4) is a composition series of length r through $\mathfrak{B}$ after passing to the appropriate quotient. Thus $s = r$. The isomorphism under 2α gives agreement between (3) and (4), and hence between (1) and (2). This proves the Jordan–Hölder theorem.

2γ. Preliminary remark. Let $\mathfrak{A}, \mathfrak{B}, \mathfrak{H}$ be modules of $\mathfrak{G}$, and suppose $\mathfrak{B}$ is simple in $\mathfrak{A}$. Then the modules $(\mathfrak{B}, \mathfrak{H})$ and $(\mathfrak{A}, \mathfrak{H})$ either coincide, or $(\mathfrak{B}, \mathfrak{H})$ is simple in $(\mathfrak{A}, \mathfrak{H})$. By the second isomorphism theorem,

$$(\mathfrak{A}, \mathfrak{B}, \mathfrak{H})/(\mathfrak{B}, \mathfrak{H}) \cong \mathfrak{A}/[\mathfrak{A}, (\mathfrak{B}, \mathfrak{H})].$$

Since $\mathfrak{B} \equiv 0 \pmod{\mathfrak{A}}$, this is

$$(\mathfrak{A}, \mathfrak{H})/(\mathfrak{B}, \mathfrak{H}) \cong \mathfrak{A}/\mathfrak{C}, \qquad \mathfrak{C} = [\mathfrak{A}, (\mathfrak{B}, \mathfrak{H})].$$

[34] Compare Masazo Sono (note 21), *On Congruences I*, §§11–13, for the case of rings. There the analogue of a composition series in noncommutative groups is also treated: series $\mathfrak{R}, \mathfrak{U}_1, \ldots, \mathfrak{U}_r, \mathfrak{E}$, where $\mathfrak{U}_i$ is an ideal and simple in $\mathfrak{U}_{i-1}$, without necessarily being an ideal in $\mathfrak{R}$. The present arguments remain valid for noncommutative groups if, by a multiple chain, one means a chain in which each $\mathfrak{U}_i$ is normal in $\mathfrak{U}_{i-1}$, and divisor chains are defined correspondingly. Compare also Dedekind, "Über die von drei Moduln erzeugte Dualgruppe," Math. Ann. 53 (1900), pp. 371–403. There, for much more general domains, the Jordan–Hölder theorem is likewise proved under the existence assumption alone; in place of the second isomorphism theorem there occurs a somewhat weaker correspondence relation, and consequently the statement of the theorem is also somewhat weaker. The principle of the proof is identical with the one given here.

Here $\mathfrak{C} \equiv 0 \pmod{\mathfrak{A}}$; on the other hand $\mathfrak{B} \equiv 0 \pmod{\mathfrak{C}}$, because $\mathfrak{B} \equiv 0 \pmod{\mathfrak{A}}$ and $\mathfrak{B} \equiv 0 \pmod{(\mathfrak{B}, \mathfrak{H})}$. Hence $\mathfrak{C}$ lies between $\mathfrak{A}$ and $\mathfrak{B}$ and is therefore, by assumption, identical with $\mathfrak{A}$ or $\mathfrak{B}$. This proves the preliminary remark.

Now let

$$\mathfrak{G}, \mathfrak{U}, \mathfrak{U}_1, \ldots, \mathfrak{U}_r, \mathfrak{E}$$

be a composition series of $\mathfrak{G}$, and let $\mathfrak{H}$ be an arbitrary module of $\mathfrak{G}$ different from $\mathfrak{G}$. To show that a composition series passes through $\mathfrak{H}$, form

$$\mathfrak{G}, (\mathfrak{U}, \mathfrak{H}), (\mathfrak{U}_1, \mathfrak{H}), \ldots, (\mathfrak{U}_r, \mathfrak{H}), \mathfrak{H}.$$

By the preliminary remark, the distinct modules among these form a composition series from $\mathfrak{G}$ to $\mathfrak{H}$. Thus the first proper term is simple in $\mathfrak{G}$ (this is the special case 2α when it coincides with $\mathfrak{B}$). By 2α there is in $\mathfrak{G}$ a composition series of length $r < r + 1$ through that term, and consequently, by the induction hypothesis, a composition series through $\mathfrak{H}$; adjoining $\mathfrak{G}$ gives a composition series of $\mathfrak{G}$ through $\mathfrak{H}$. Repeating the argument shows that such a series may in particular be chosen as an extension of the series just constructed from $\mathfrak{G}$ to $\mathfrak{H}$.

2δ. Proof of the multiple-chain condition. Suppose again that a composition series of length $r + 1$ exists in $\mathfrak{G}$, and let

$$\mathfrak{G}, \mathfrak{B}, \mathfrak{B}_1, \ldots, \mathfrak{B}_i, \ldots$$

be a multiple chain, each term being a proper multiple of the preceding one. That the chain begins with $\mathfrak{G}$ is no restriction of generality, since $\mathfrak{G}$ may always be adjoined as first term. By 2γ there is a composition series through $\mathfrak{B}$, and hence $\mathfrak{B}$ has a composition series of shorter length. By the induction hypothesis, the chain

$$\mathfrak{B}, \mathfrak{B}_1, \ldots, \mathfrak{B}_i,$$

terminates after finitely many steps. Hence the same is true after adjoining $\mathfrak{G}$.

2ε. Proof of the divisor-chain condition. Again assume a composition series of length $r + 1$ in $\mathfrak{G}$, and let

$$\mathfrak{G}, \mathfrak{C}, \mathfrak{C}_1, \ldots, \mathfrak{C}_i,$$

be a divisor chain, each term being a proper divisor of the preceding one. Again there is no restriction in assuming that the chain begins with $\mathfrak{G}$. By 2γ there is a composition series through $\mathfrak{C}$, and hence in $\mathfrak{G}/\mathfrak{C}$ there is a composition series of shorter length. By the induction hypothesis the divisor chain

$$\mathfrak{G}/\mathfrak{C},\ \mathfrak{C}_1/\mathfrak{C}, \ldots, \mathfrak{C}_i/\mathfrak{C},$$

terminates after finitely many steps. By the one-to-one correspondence furnished by the first isomorphism theorem (§4, 2), the same holds for the original chain beginning with $\mathfrak{G}$, and hence also for the chain with $\mathfrak{E}$ adjoined. The equivalence of the two assumptions – existence of a composition series and the double-chain condition – is thereby proved.

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